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Structural properties of graph products

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Abstract

Dujmović, Joret, Micek, Morin, Ueckerdt, and Wood established that every planar graph is a subgraph of the strong product of a graph with bounded treewidth and a path. Motivated by this result, this paper systematically studies various structural properties of cartesian, direct and strong products. In particular, we characterise when these graph products contain a given complete multipartite subgraph, determine tight bounds for their degeneracy, establish new lower bounds for the treewidth of cartesian and strong products, and characterise when they have bounded treewidth and when they have bounded pathwidth.

KEYWORDS

degeneracy, graph products, treewidth

1 | INTRODUCTION

Dujmović et al. [25] recently showed that every planar graph is a subgraph of the strong product of a graph with bounded treewidth and a path.

Theorem 1 (Dujmović et al. [25]). *Every planar graph is isomorphic to a subgraph of:*

- $H \boxtimes P$ for some graph H of treewidth at most 8 and for some path P ;
- $H \boxtimes P \boxtimes K_3$ for some graph H of treewidth at most 3 and for some path P .

This breakthrough has been the key tool to resolve several major open problems regarding queue layouts [25], nonrepetitive colourings [24], centred colourings [17], and adjacency

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labelling schemes [7, 23, 31]. Treewidth measures how similar a graph is to a tree and is an important parameter in algorithmic and structural graph theory (see [4, 38, 59]). Graphs with bounded treewidth are a simpler class of graphs compared to planar graphs. Theorem 1 therefore reduces problems on a complicated class of graphs (planar graphs) to a simpler class of graphs (bounded treewidth). Ueckerdt et al. [71] improved Theorem 1 by replacing ‘treewidth at most 8’ with ‘simple treewidth at most 6’. Analogous *graph product structure theorems* hold for other graph classes such as graphs with bounded genus [19, 25], minor-closed graph classes [25], map graphs [26], squaregraphs [39], k -planar graphs [26], fan-planar [40], k -fan bundle planar [40] and infinite graphs [43].

These results motivate further study into the structural properties of graph products. In this paper, we explore the following properties of cartesian, direct and strong products: complete multipartite subgraphs, degeneracy, pathwidth and treewidth. Our key contributions are the following:

Complete multipartite subgraphs: We characterise the presence of complete multipartite subgraphs in cartesian, direct and strong products. These results are presented in Section 2.

Degeneracy: A graph G is d -degenerate if every subgraph of G has minimum degree at most d . The *degeneracy* of G , $\text{degen}(G)$, is the minimum integer d such that G is d -degenerate. Degeneracy is an important graph parameter as it measures the sparsity of a graph. Moreover, d -degenerate graphs are $(d + 1)$ -colourable and in fact, $(d + 1)$ -choosable. We present tight upper and lower bounds on the degeneracy of direct and strong products. See Section 3 for these results.

Pathwidth and treewidth of cartesian and strong products: We establish two new lower bounds for the treewidth of cartesian and strong products. The first bound states that if one graph does not admit small ϵ -separations and the other graph is connected with many vertices, then the cartesian product has large treewidth. The second bound states that if one graph has large treewidth and the other graph has large Hadwiger number, then the strong product has large treewidth. In addition, we characterise when the cartesian and strong product of two monotone graph classes has bounded treewidth and when it has bounded pathwidth. These results are presented in Section 4.1.

Pathwidth and treewidth of direct products: We characterise when the direct product of two monotone graph classes has bounded treewidth and when it has bounded pathwidth. For treewidth, the characterisation states that the direct product of two graph classes has bounded treewidth if and only if the connected graphs in one of the classes have a bounded number of vertices while the graphs in the other class have bounded treewidth; or if the connected graphs in one of the classes have bounded vertex cover number while the graphs in the other class have bounded treewidth and bounded maximum degree. For pathwidth, our characterisation is directly analogous to that for treewidth with the stronger condition that the second class has bounded pathwidth. We also demonstrate that the treewidth of a graph is polynomially tied to the treewidth of the direct product of the graph with K_2 . To our knowledge, it was previously open whether these two parameters were tied. These results are presented in Sections 4.2 and 4.3.

This line of research has been previously explored for the following properties of graph products: connectivity [11, 12, 74, 78]; queue-number [75]; stack-number [22, 29, 42, 51, 58]; thinness [9]; boxicity and cubicity [13]; polynomial growth [27]; bounded expansion and colouring numbers [27]; chromatic number [21, 45, 63, 66, 67, 72, 73, 80]; and Hadwiger number [1, 15, 44, 49, 54, 57, 76, 79]. See the handbook by Hammack, Imrich, and Klavžar [34] for an in-depth treatment of graph products.

1.1 | Preliminaries

For every integer $k \geq 1$, let $[k] := \{1, \dots, k\}$. A *graph parameter* is a function β such that $\beta(G) \in \mathbb{R}$ for every graph G and $\beta(G_1) = \beta(G_2)$ for all isomorphic graphs G_1 and G_2 . Two graph parameters α and β are *tied* if there exists a function f such that $\alpha(G) \leq f(\beta(G))$ and $\beta(G) \leq f(\alpha(G))$. If the function f is polynomial, then they are *polynomially tied*.

Let H and G be graphs. H is a *minor* of G if a graph isomorphic to H can be obtained from G by vertex deletion, edge deletion, and edge contraction. A *model* of H in G is a function μ from $V(H)$ to disjoint connected subgraphs in G such that $\mu(u)$ and $\mu(v)$ are adjacent whenever $uv \in E(H)$. It is folklore that H is a minor of G if and only if G contains a model of H . Let $v(G) := |V(G)|$ be the *order* of G . Let $\tilde{v}(G)$ be the maximum order of a connected component of G . For $v \in V(G)$, let $N_G(v) := \{u \in V(G) : uv \in E(G)\}$ and $N_G[v] := N(v) \cup \{v\}$. When the graph G is clear, we drop the subscript G and use the notation $N(v)$ and $N[v]$. Let $\Delta(G)$ and $\delta(G)$, respectively denote the maximum and minimum degree of G .

A *class* of graphs $\mathcal{G}$ is a set of graphs that is closed under isomorphism and contains a graph with a nonempty vertex set. $\mathcal{G}$ is *hereditary* if it is closed under induced subgraphs, *monotone* if it is closed under subgraphs, and *minor-closed* if it is closed under minors. For a graph parameter β , let $\beta(\mathcal{G}) := \sup\{\beta(G) : G \in \mathcal{G}\}$. We say that $\beta(\mathcal{G})$ is *bounded* if there exists $c \in \mathbb{R}$ such that $\beta(G) \leq c$ for every graph $G \in \mathcal{G}$, otherwise we say that $\beta(\mathcal{G})$ is *unbounded*.

Let G_1 and G_2 be graphs. A *graph product* $G_1 \bullet G_2$ is defined with vertex set:

$$V(G_1 \bullet G_2) := \{(a, v) : a \in V(G_1), v \in V(G_2)\}.$$

The *cartesian product* $G_1 \square G_2$ consists of edges of the form $(a, v)(b, u)$, where either $ab \in E(G_1)$ and $v = u$, or $uv \in E(G_2)$ and $a = b$. The *direct product* $G_1 \times G_2$ consists of edges of the form $(a, v)(b, u)$ where $ab \in E(G_1)$ and $uv \in E(G_2)$. This product is also known as the *tensor product*, the *Kronecker product* and the *cross product*. The *strong product* $G_1 \boxtimes G_2$ is defined as $(G_1 \square G_2) \cup (G_1 \times G_2)$. For a graph product $\bullet \in \{\square, \times, \boxtimes\}$, graph classes $\mathcal{G}_1$ and $\mathcal{G}_2$, and graph H , let $\mathcal{G}_1 \bullet \mathcal{G}_2$ be the graph class $\{G_1 \bullet G_2 : G_1 \in \mathcal{G}_1, G_2 \in \mathcal{G}_2\}$ and let $\mathcal{G}_1 \bullet H$ be the graph class $\{G_1 \bullet H : G_1 \in \mathcal{G}_1\}$.

The following well-known properties of graph products are a straightforward consequence of their definition:

- for all $\bullet \in \{\square, \times, \boxtimes\}$ and graphs G_1 and G_2 with subgraphs $H_1 \subseteq G_1$ and $H_2 \subseteq G_2$, we have $H_1 \bullet H_2 \subseteq G_1 \bullet G_2$;
- for all hereditary graph classes $\mathcal{G}_1$ and $\mathcal{G}_2$, we have $\mathcal{G}_1 \subseteq \mathcal{G}_1 \square \mathcal{G}_2$ and $\mathcal{G}_2 \subseteq \mathcal{G}_1 \square \mathcal{G}_2$;
- for all graphs G_1 and G_2 and vertices $v_i \in V(G_i)$, where $\deg_{G_i}(v_i) = d_i$ for $i \in \{1, 2\}$:
 - $\deg_{G_1 \square G_2}((v_1, v_2)) = d_1 + d_2$;
 - $\deg_{G_1 \times G_2}((v_1, v_2)) = d_1 d_2$; and
 - $\deg_{G_1 \boxtimes G_2}((v_1, v_2)) = d_1 + d_2 + d_1 d_2$.

For a subgraph $Z \subseteq G_1 \bullet G_2$ of a graph product where $\bullet \in \{\square, \times, \boxtimes\}$, the *projection of Z onto G_1* is the subgraph of G_1 induced by the set of vertices $u_1 \in V(G_1)$ such that $(u_1, u_2) \in V(Z)$ for some $u_2 \in V(G_2)$, and the *projection of Z onto G_2* is the subgraph of G_2 induced by the set of vertices $v_2 \in V(G_2)$ such that $(v_1, v_2) \in V(Z)$ for some $v_1 \in V(G_1)$.

2 | COMPLETE MULTIPARTITE SUBGRAPHS

For integers $d \geq 2$ and $n_1, \dots, n_d \geq 1$, a *complete d -partite graph* $K_{n_1, \dots, n_d}$ has a partition of its vertex set, $(A_1, \dots, A_d)$, such that for all distinct $i, j \in [d]$ and $a_i \in A_i$ and $a_j \in A_j$, we have $a_i a_j \in E(K_{n_1, \dots, n_d})$ and $|A_i| = n_i$. When $d = 2$, it is a *complete bipartite graph*. Observe that the graph $K_1, \dots, 1$ is a complete graph. In this section, we characterise when a cartesian, direct and strong product contains a given complete multipartite subgraph.

2.1 | Cartesian product

We begin with cartesian products. Let G_1 and G_2 be graphs. We say that $(u_1, v_1), \dots, (u_d, v_d) \in V(G_1 \square G_2)$ are *aligned* if $u_1 = u_2 = \dots = u_d$ or $v_1 = v_2 = \dots = v_d$. Observe that for a subgraph $H \subseteq G_1 \square G_2$, if the vertex set $V(H)$ is aligned, then H is isomorphic to a subgraph of G_1 or G_2 . For a set $X \subseteq V(G_1 \square G_2)$, if every pair of vertices in X is aligned, then X is aligned. Furthermore, if x and y are adjacent vertices in $G_1 \square G_2$, then x, y are aligned. Hence, we have the following.

Lemma 2. *For all graphs, G_1 and G_2 , every clique in $G_1 \square G_2$ is aligned.*

The next two lemmas are used in our characterisation of complete multipartite subgraphs in a cartesian product.

Lemma 3. *For all graphs G_1 and G_2 , if $X \subseteq V(G_1 \square G_2)$ and $|\bigcap_{x \in X} N(x)| > 2$, then X is aligned.*

Proof. Let $X := \{(u_1, v_1), \dots, (u_k, v_k)\}$ for some integer $k \geq 1$. If $k = 1$ then the claim holds trivially. So assume that $k \geq 2$. For the sake of contradiction, suppose there exists distinct $i, j \in [k]$ such that $u_i \neq u_j$ and $v_i \neq v_j$. Then

$$\begin{aligned} N((u_i, v_i)) \cap N((u_j, v_j)) &= (\{(u_i, v) : vv_i \in E(G_2)\} \cup \{(u, v_i) : uu_i \in E(G_1)\}) \\ &\quad \cap (\{(u_j, v) : vv_j \in E(G_2)\} \cup \{(u, v_j) : uu_j \in E(G_1)\}) \\ &\subseteq \{(u_i, v_j), (u_j, v_i)\}. \end{aligned}$$

But this contradicts the assumption that the intersection of the neighbourhoods of (u_i, v_i) and (u_j, v_j) has a size greater than 2. Thus, every pair of vertices in X is aligned and hence, X is aligned. $\square$

Lemma 4. *If $(u_1, v_1), (u_2, v_2)$ are distinct vertices that are aligned in $V(G_1 \square G_2)$, then $N[(u_1, v_1)] \cap N[(u_2, v_2)]$ is aligned.*

Proof. Without loss of generality, $v_1 = v_2 = v^*$. Then

$$\begin{aligned} N((u_1, v^*)) \cap N((u_2, v^*)) &= (\{(u_1, v) : vv^* \in E(G_2)\} \cup \{(u, v^*) : uu_1 \in E(G_1)\}) \\ &\quad \cap (\{(u_2, v) : vv^* \in E(G_2)\} \cup \{(u, v^*) : uu_2 \in E(G_1)\}) \\ &\subseteq \{(x, v^*) : x \in V(G_1)\}. \end{aligned}$$

Hence $N[(u_1, v^*)] \cap N[(u_2, v^*)]$ is aligned. $\square$

We now characterise when a cartesian product contains a given complete multipartite subgraph.

Theorem 5. *For all integers $d \geq 2$ and $n_1, \dots, n_d \geq 1$, and for all graphs G_1 and G_2 with nonempty vertex sets and maximum degree Δ_1 and Δ_2 , respectively, $K_{n_1, \dots, n_d} \subseteq G_1 \square G_2$ if and only if at least one of the following conditions hold:*

- $K_{n_1, \dots, n_d}$ is a subgraph of G_1 or G_2 ;
- $d = 2$ and $(n_1, n_2) = (2, 2)$ and $K_2 \subseteq G_1$ and $K_2 \subseteq G_2$; or
- $d = 2$ and $(n_1, n_2) = (1, s)$ for some integer $s \geq 1$ and $\Delta_1 + \Delta_2 \geq s$.

Proof. Clearly if $K_{n_1, \dots, n_d}$ is a subgraph of either G_1 or G_2 , then it is a subgraph of $G_1 \square G_2$.

Observe that $K_{1,s}$ is a subgraph of a graph G if and only if the maximum degree of G is at least s . As such, $K_{1,s}$ is a subgraph of $G_1 \square G_2$ if and only if $\Delta_1 + \Delta_2 \geq s$. Now consider the bipartite graph $K_{2,2}$. Suppose $xy \in E(G_1)$ and $uv \in E(G_2)$. Let $A = \{(x, u), (y, v)\}$ and $B = \{(y, u), (x, v)\}$. Then (A, B) defines a $K_{2,2}$ subgraph in $G_1 \square G_2$. Now suppose that $K_{2,2} \subseteq G_1 \square G_2$ and K_2 is not a subgraph of G_1 . Then $E(G_1) = \emptyset$ and hence $V(G_1)$ is an independent set. As such, the connected components of $G_1 \square G_2$ are isomorphic to the connected components of G_2 . Since $K_{2,2}$ is connected, it is a subgraph of G_2 , as required.

It remains to consider the cases when $(n_1, \dots, n_d)$ is not equal to $(1, s)$ or $(2, 2)$. First, suppose $d \geq 3$ and that $K_{n_1, \dots, n_d} \subseteq G_1 \square G_2$ with partition $(A_1, \dots, A_d)$. For every $i \in [d]$, let $(x_i, y_i) \in A_i$. Then $X := \{(x_1, y_1), \dots, (x_d, y_d)\}$ induces a K_d subgraph in $G_1 \square G_2$. By Lemma 2, X is aligned. Furthermore, for every $(u, v) \in V(K_{n_1, \dots, n_d})$, there exists distinct $i, j \in [d]$ such that $(u, v) \in N[(x_i, y_i)] \cap N[(x_j, y_j)]$. By Lemma 4, $V(K_{n_1, \dots, n_d})$ is aligned and hence $K_{n_1, \dots, n_d}$ is a subgraph of G_1 or G_2 , as required.

Now suppose $d = 2$. Let (A_1, A_2) be the bipartition of K_{n_1, n_2} where $n_1 = |A_1| \geq 2$ and $n_2 = |A_2| \geq 3$. By Lemma 3, A_1 is aligned. By Lemma 4, $V(K_{n_1, n_2})$ is aligned and hence K_{n_1, n_2} is a subgraph of G_1 or G_2 . $\square$

2.2 | Direct product

The next theorem characterises when a direct product contains a given complete multipartite subgraph.

Theorem 6. *For all integers $d \geq 2$ and $n_1, \dots, n_d \geq 1$, and for all graphs G_1 and G_2 with nonempty edge sets, $K_{n_1, \dots, n_d} \subseteq G_1 \times G_2$ if and only if $K_{a_1, \dots, a_d} \subseteq G_1$ and $K_{b_1, \dots, b_d} \subseteq G_2$, where a_i, b_i are positive integers and $n_i \leq a_i b_i$ for all $i \in [d]$.*

Proof. Suppose $K_{a_1, \dots, a_d} \subseteq G_1$ and $K_{b_1, \dots, b_d} \subseteq G_2$. Let $(A_1, \dots, A_d)$ be the partition of $K_{a_1, \dots, a_d}$ and $(B_1, \dots, B_d)$ be the partition of $K_{b_1, \dots, b_d}$. Consider the graph $K_{a_1, \dots, a_d} \times K_{b_1, \dots, b_d}$. For all distinct $i, j \in [d]$, every vertex $v_i \in (A_i, B_i)$ is adjacent to every vertex $v_j \in (A_j, B_j)$. Hence

$((A_1, B_1), \dots, (A_d, B_d))$ defines a $K_{a_1 b_1, \dots, a_d b_d}$ subgraph in $K_{a_1, \dots, a_d} \times K_{b_1, \dots, b_d}$. Since $n_i \leq a_i b_i$ for all $i \in [d]$, we have $K_{n_1, \dots, n_d} \subseteq G_1 \times G_2$.

For the second direction, suppose $K_{n_1, \dots, n_d} \subseteq G_1 \times G_2$. Let $(N_1, \dots, N_d)$ correspond to the partition of $K_{n_1, \dots, n_d}$. For $i \in \{1, 2\}$ and $j \in [d]$, let $N_j^{(i)}$ be the vertices in N_j projected onto G_i . By the definition of the direct product, $\{(x, v_2) : x \in V(G_1)\}$ and $\{(v_1, y) : y \in V(G_2)\}$ are independent sets for all $v_1 \in V(G_1)$ and $v_2 \in V(G_2)$. As such, $N_j^{(i)}$ and $N_k^{(i)}$ are disjoint for distinct $j, k \in [d]$ and $i \in \{1, 2\}$. Moreover, as every vertex in N_j is adjacent to every vertex in N_k , it follows that $K_{a_1, \dots, a_d} \subseteq G_1[N_1^{(1)} \cup \dots \cup N_d^{(1)}]$ and $K_{b_1, \dots, b_d} \subseteq G_2[N_1^{(2)} \cup \dots \cup N_d^{(2)}]$, where $a_j = |N_j^{(1)}|$, and $b_j = |N_j^{(2)}|$ for all $j \in [d]$. Since there are $a_j b_j$ vertices in $(N_j^{(1)}, N_j^{(2)})$, we have $n_j \leq a_j b_j$ for all $j \in [d]$, as required. $\square$

2.3 | Strong product

Let $K_{n_1, \dots, n_d, \bar{x}}$ be the graph obtained from the complete multipartite graph $K_{n_1, \dots, n_d, x}$ by adding an edge between each pair of vertices in the part of size $x \geq 0$. More formally, $V(K_{a_1, \dots, a_d, \bar{x}}) = A_1 \cup \dots \cup A_d \cup X$, such that $A_1, \dots, A_d, X$ are pairwise disjoint sets where for distinct $j, k \in [d]$, we have $|A_j| = a_j$, $|X| = x$, and $uv, vw_1, w_1 w_2 \in E(K_{a_1, \dots, a_d, \bar{x}})$ where $u \in A_j$, $v \in A_k$, and $w_1, w_2 \in X$. Observe that $K_{n_1, \dots, n_d, \bar{x}} = K_{n_1, \dots, n_d, 1, \dots, 1}$ and $K_{n+c} = K_{1, \dots, 1, \bar{c}}$.

Lemma 7. *If $K_{a_1, \dots, a_d, \bar{x}} \subseteq G_1$ and $K_{b_1, \dots, b_d, \bar{y}} \subseteq G_2$, then*

$$K_{a_1 b_1 + a_1 y + b_1 x, \dots, a_d b_d + a_d y + b_d x, \bar{xy}} \subseteq G_1 \boxtimes G_2.$$

Proof. Let $A_1, \dots, A_d, X$ be the subsets of $V(G_1)$ defining a $K_{a_1, \dots, a_d, \bar{x}}$ subgraph of G_1 . Let $B_1, \dots, B_d, Y$ be the subsets of $V(G_2)$ defining a $K_{b_1, \dots, b_d, \bar{y}}$ subgraph of G_2 . Then $(A_1 \times B_1) \cup (A_1 \times Y) \cup (B_1 \times X), \dots, (A_d \times B_d) \cup (A_d \times Y) \cup (B_d \times X), (X \times Y)$ define a $K_{a_1 b_1 + a_1 y + b_1 x, \dots, a_d b_d + a_d y + b_d x, \bar{xy}}$ subgraph in $G_1 \boxtimes G_2$. $\square$

The next theorem characterises when a strong product contains a given complete multipartite subgraph.

Theorem 8. *For all integers $d \geq 2$ and $n_1, \dots, n_d \geq 1$, and for all graphs G_1 and G_2 , we have $K_{n_1, \dots, n_d} \subseteq G_1 \boxtimes G_2$ if and only if there exists nonnegative integers $a_1, \dots, a_d, b_1, \dots, b_d, z_1, \dots, z_d, x, y$ such that:*

- $K_{a_1, \dots, a_d, \bar{x}} \subseteq G_1$;
- $K_{b_1, \dots, b_d, \bar{y}} \subseteq G_2$;
- $n_j \leq a_j b_j + a_j y + b_j x + z_j$ for all $j \in [d]$; and
- $z_1 + \dots + z_d \leq xy$.

Proof. First, suppose that $K_{n_1, \dots, n_d}$ is a subgraph in $G_1 \boxtimes G_2$. Let $(N_1, \dots, N_d)$ be the partition of the $K_{n_1, \dots, n_d}$ subgraph. For all $i \in \{1, 2\}$ and $j \in [d]$, let $N_j^{(i)}$ be the vertex set for the projection of $G_1 \boxtimes G_2[N_j]$ onto G_i , let $\tilde{N}_j^{(i)} := N_j^{(i)} \setminus (\bigcup_{h \in [d], h \neq j} N_h^{(i)})$ and let

$Z^{(i)} := \bigcup_{j \in [d]} (N_j^{(i)} \setminus \tilde{N}_j^{(i)})$. Then $(\tilde{N}_1^{(i)}, \dots, \tilde{N}_d^{(i)}, Z^{(i)})$ are the colour classes of a complete multipartite subgraph of G_i . For all $j \in [d]$, let $a_j := |\tilde{N}_j^{(1)}|$, $b_j := |\tilde{N}_j^{(2)}|$, $x := |Z^{(1)}|$ and $y := |Z^{(2)}|$. Thus $K_{a_1, \dots, a_d, x} \subseteq G_1$ and $K_{b_1, \dots, b_d, y} \subseteq G_2$. Consider distinct vertices $u_1, v_1 \in Z^{(1)}$. Then there exists distinct $j, k \in [d]$ such that $(u_1, u_2) \in N_j$ and $(v_1, v_2) \in N_k$ for some $u_2, v_2 \in V(G_2)$ which implies that $u_1 v_1 \in E(G_1)$. Similarly, the vertices in $Z^{(2)}$ are also pairwise adjacent in G_2 . Hence, $K_{a_1, \dots, a_d, \bar{x}} \subseteq G_1$ and $K_{b_1, \dots, b_d, \bar{y}} \subseteq G_2$. Let $Z := Z^{(1)} \times Z^{(2)}$ and $z_j := |Z \cap N_j|$ for all $j \in [d]$. Then $z_1 + \dots + z_d \leq |Z| = xy$. Since $N_j \subseteq (\tilde{N}_j^{(1)} \times N_j^{(2)}) \cup (\tilde{N}_j^{(1)} \times Z_2) \cup (Z_1 \cap N_j^{(2)})$ we have $n_j \leq a_j b_j + a_j y + b_j x + z_j$ for all $j \in [d]$ as required.

Now suppose that $K_{a_1, \dots, a_d, \bar{x}} \subseteq G_1$ and $K_{b_1, \dots, b_d, \bar{y}}$ where for all $j \in [d]$, a_j, b_j, x, y are nonnegative integers. By Lemma 7, we have $K_{n_1, \dots, n_d, \bar{xy}} \subseteq G_1 \boxtimes G_2$. Splitting the colour class of size xy into sets of size $z_1, \dots, z_d$, and combining z_j with the j th colour classes for all j , we obtain a $K_{a_1 b_1 + a_1 y + b_1 x + z_1, \dots, a_d b_d + a_d y + b_d x + z_d}$ subgraph in $G_1 \boxtimes G_2$. Since $n_j \leq a_j b_j + a_j y + b_j x + z_j$ for all $j \in [d]$, we have $K_{n_1, \dots, n_d} \subseteq G_1 \boxtimes G_2$. $\square$

For a graph G , let $\omega(G)$ be the maximum size of a clique in G . To illustrate the application of Theorem 8, we show how it implies the following well-known result.

Corollary 9. *For all graphs G and H , $\omega(G \boxtimes H) = \omega(G)\omega(H)$.*

Proof. Let $c := \omega(G)$ and $d := \omega(H)$. Since $K_{0, \dots, 0, \bar{c}} \subseteq G$ and $K_{0, \dots, 0, \bar{d}} \subseteq H$, by Theorem 8 we have $K_{1, \dots, 1} = K_{cd} \subseteq G \boxtimes H$ by setting $z_i := 1$ for all $i \in [cd]$. Hence, $\omega(G \boxtimes H) \geq \omega(G)\omega(H)$. Now suppose that $K_n = K_{1, \dots, 1} \subseteq G \boxtimes H$. By Theorem 8, there exists nonnegative integers $a_1, \dots, a_n, b_1, \dots, b_n, z_1, \dots, z_n, x, y$ such that $K_{a_1, \dots, a_n, \bar{x}} \subseteq G_1$; and $K_{b_1, \dots, b_n, \bar{y}} \subseteq G_2$; and:

$$n_i \leq a_i b_i + a_i y + b_i x + z_i \quad \text{for all } i \in [n]; \text{ and} \quad (1)$$

$$z_1 + \dots + z_n \leq xy. \quad (2)$$

Let $A := \{i \in [n] : a_i \geq 1\}$, $B := \{i \in [n] : b_i \geq 1\}$ and $Z := \{i \in [n] : z_i \geq 1\}$. Then $|A| + |B| + |Z| \geq n$ since by Equation (1), a_i, b_i or z_i is at least 1 for all $i \in [n]$. By Equation (2), $|Z| \leq xy$. Now $K_{1, \dots, 1, \bar{x}} = K_c \subseteq G$, and $K_{1, \dots, 1, \bar{y}} = K_d \subseteq H$, where $c := |A| + x$ and $d := |B| + y$. Therefore, $cd = (|A| + x)(|B| + y) \geq |A| + |B| + xy \geq n$. Hence $\omega(G)\omega(H) \geq \omega(G \boxtimes H)$, as required. $\square$

The case of complete bipartite subgraphs is of particular interest. The following result generalises a lemma due to Bonnet, Geniet, Kim, Thomassé, and Watrigant [8, lemma 7.2].

Corollary 10. *For all integers $t \geq s \geq 1$ and $\Delta \geq 1$, for all graphs G and H where G is $K_{s,t}$ -free and H has maximum degree Δ , $G \boxtimes H$ is $K_{(s-1)(\Delta+1)+1, (s+t)(\Delta+1)}$ -free. Moreover, for every integer $n \geq 1$, there exists a $K_{s,t}$ -free graph $\tilde{G}$ and a graph $\tilde{H}$ with maximum degree Δ such that $K_{(s-1)(\Delta+1), n} \subseteq \tilde{G} \boxtimes \tilde{H}$.*

Proof. For the sake of contradiction, suppose that $K_{(s-1)(\Delta+1)+1, (s+t)(\Delta+1)} \subseteq G \boxtimes H$. By Theorem 8, there exists non-negative integers $a_1, a_2, b_1, b_2, z_1, z_2, x, y$ such that $K_{a_1, a_2, \bar{x}} \subseteq G_1$; and $K_{b_1, b_2, \bar{y}} \subseteq G_2$; and

$$(s-1)(\Delta+1)+1 \leq a_1(b_1+y) + b_1x + z_1; \quad (3)$$

$$(s+t)(\Delta+1) \leq a_2(b_2+y) + b_2x + z_2; \text{ and} \quad (4)$$

$$z_1 + z_2 \leq xy. \quad (5)$$

Observe that if b_1 and y are both equal to 0 then Equations (3) and (5) cannot both be satisfied. As such, $b_2 + y \leq \Delta + 1$ since H has maximum degree Δ . Similarly, b_2 and y cannot both equal to 0 as this will violate Equation (4). Thus, $b_1 + y \leq \Delta + 1$. By Equation (5), $z_1 \leq xy$ and $z_2 \leq xy$. By Equation (3), we have $(\Delta+1)(s-1)+1 \leq a_1(b_1+y) + b_1x + xy \leq (a_1+x)(\Delta+1)$. As such, $a_1+x > s-1$. Similarly, by Equation (4), $(s+t)(\Delta+1) \leq a_2(b_2+y) + b_2x + xy \leq (a_2+x)(\Delta+1)$. As such, $a_2+x \geq s+t$. This forces $K_{s,t} \subseteq G$, a contradiction.

Finally, let $\tilde{G} := K_{s-2, n_1}$ and $\tilde{H} := K_{\Delta, 0, 1}$. Then $\tilde{G}$ is $K_{s,t}$ -free and H has maximum degree Δ . By Theorem 8, $K_{n_1, n_2} \subseteq \tilde{G} \boxtimes \tilde{H}$, where $n_1 = \Delta(s-2) + (s-2) + \Delta + 1 = (s-1)(\Delta+1)$ and $n_2 = n$. $\square$

3 | DEGENERACY

Recall that the *degeneracy* of a graph G , $\text{degen}(G)$, is the minimum integer d such that every subgraph of G has minimum degree at most d . Equivalently, the *degeneracy* of G is the maximum, taken over all subgraphs H of G , of the minimum degree of H .

Bickle [2] determined the degeneracy of a cartesian product. In particular, for all graphs G_1 and G_2 , $\text{degen}(G_1 \square G_2) = \text{degen}(G_1) + \text{degen}(G_2)$.

3.1 | Direct product

We now prove tight bounds for the degeneracy of a direct product. We make use of the following observation.

Observation 11. For all integers $t_i \geq s_i \geq 1$, where $i \in \{1, 2\}$,

$$\text{degen}(K_{s_1, t_1} \times K_{s_2, t_2}) = \min\{s_1 t_2, s_2 t_1\}.$$

This follows from the fact that $K_{s_1, t_1} \times K_{s_2, t_2}$ is the disjoint union of $K_{s_1 t_2, s_2 t_1}$ and $K_{s_1 s_2, t_1 t_2}$, and that $\text{degen}(K_{s,t}) = \min\{s, t\}$.

Theorem 12. For $i \in \{1, 2\}$, let G_i be a graph with maximum degree Δ_i and $\text{degen}(G_i) = d_i$ that contains K_{s_i, t_i} as a subgraph where $s_i \leq t_i$. Then:

$$\max\{d_1 d_2, \min\{s_1 t_2, s_2 t_1\}, \min\{\Delta_1, \Delta_2\}\} \leq \text{degen}(G_1 \times G_2) \leq \min\{d_1 \Delta_2, d_2 \Delta_1\}.$$

Proof. We first prove the lower bound. Let Q_i be a subgraph of G_i with $\delta(Q_i) = d_i$. Then $\delta(Q_1 \times Q_2) = d_1 d_2$ and thus, $\text{degen}(G_1 \times G_2) \geq d_1 d_2$. Furthermore, since G_i contains K_{1, Δ_i} and K_{s_i, t_i} as a subgraph, by Observation 11, $\text{degen}(G_1 \times G_2) \geq \min\{s_1 t_2, s_2 t_1\}$ and $\text{degen}(G_1 \times G_2) \geq \min\{\Delta_1, \Delta_2\}$.

Now we prove the upper bound. Without loss of generality, $d_1 \Delta_2 \leq d_2 \Delta_1$. Let Z be a subgraph of $G_1 \times G_2$. Our goal is to show that $\delta(Z) \leq d_1 \Delta_2$. Let X be the projection of Z onto G_1 . Since G_1 is d_1 -degenerate, there exists a vertex $v_1 \in V(X)$ with $\deg_X(v_1) \leq d_1$. By construction of X , we have $(v_1, v_2) \in V(Z)$ for some $v_2 \in V(G_2)$. The neighbourhood of (v_1, v_2) in Z is a subset of $\{(u_1, u_2) : u_1 v_1 \in E(X), u_2 v_2 \in E(G_2)\}$. Now $|\{(u_1, u_2) : u_1 v_1 \in E(X), u_2 v_2 \in E(G_2)\}| \leq |\{u_1 \in V(X) : u_1 v_1 \in E(X)\}| |\{u_2 \in V(X) : u_2 v_2 \in E(G_2)\}| \leq d_1 \Delta_2$. Thus (v_1, v_2) has degree at most $d_1 \Delta_2$ in Z and hence $\delta(Z) \leq d_1 \Delta_2$ as required. $\square$

When both graphs are regular, the upper and lower bounds in Theorem 12 are equal. Furthermore, by Observation 11, the direct product of two complete bipartite graphs realises the upper bound in Theorem 12. We now show that there is a family of graphs that realises the lower bound in Theorem 12.

Lemma 13. *For all integers $d_i, \Delta_i, s_i, t_i \geq 1$ where $s_i \leq d_i \leq \Delta_i$ and $s_i \leq t_i \leq \Delta_i$ for $i \in \{1, 2\}$, there exists graphs G_1 and G_2 where for $i \in \{1, 2\}$, G_i has maximum degree Δ_i , $\text{degen}(G_i) = d_i$, and G_i contains K_{s_i, t_i} as a subgraph, such that*

$$\text{degen}(G_1 \times G_2) = \max\{d_1 d_2, \min\{s_1 t_2, s_2 t_1\}, \min\{\Delta_1, \Delta_2\}\}.$$

Proof. Let $\tilde{d} := \max\{d_1 d_2, \min\{s_1 t_2, s_2 t_1\}, \min\{\Delta_1, \Delta_2\}\}$. For $i \in \{1, 2\}$, let H_i be a d_i -regular graph and let G_i be the disjoint union of H_i , K_{s_i, t_i} and K_{1, Δ_i} . Then G_i has maximum degree Δ_i , $\text{degen}(G_i) = d_i$, and G_i contains K_{s_i, t_i} and K_{1, Δ_i} as subgraphs.

We now show that $\text{degen}(G_1 \times G_2) = \tilde{d}$. Theorem 12 provides the lower bound. For the upper bound, our goal is to show that $\text{degen}(J_1 \times J_2) \leq \tilde{d}$ whenever $J_i \in \{H_i, K_{s_i, t_i}, K_{1, \Delta_i}\}$ for $i \in \{1, 2\}$. This implies that $\text{degen}(G_1 \times G_2) \leq \tilde{d}$ since every connected subgraph of $G_1 \times G_2$ is a subgraph of $J_1 \times J_2$ for some choice of $J_i \in \{H_i, K_{s_i, t_i}, K_{1, \Delta_i}\}$ where $i \in \{1, 2\}$. We proceed by case analysis.

First, $\text{degen}(H_1 \times H_2) = d_1 d_2$ since H_1 and H_2 are regular. By Observation 11,

$$\begin{aligned} \text{degen}(K_{s_1, t_1} \times K_{s_2, t_2}) &= \min\{s_1 t_2, s_2 t_1\}, \\ \text{degen}(K_{s_1, t_1} \times K_{1, \Delta_2}) &= \min\{t_1, \Delta_2\}, \\ \text{degen}(K_{1, \Delta_1} \times K_{s_2, t_2}) &= \min\{\Delta_1, t_2\}, \text{ and} \\ \text{degen}(K_{1, \Delta_1} \times K_{1, \Delta_2}) &= \min\{\Delta_1, \Delta_2\}. \end{aligned}$$

Clearly, the degeneracy is at most $\tilde{d}$ for each of the above graphs. Now consider the graph $H_1 \times K_{s_2, t_2}$. Each vertex has degree $d_1 s_2$ or $d_1 t_2$. Furthermore, the set of vertices with degree equal to $d_1 t_2$ are an independent set. Hence, every subgraph of $H_1 \times K_{s_2, t_2}$ contains a vertex with degree at most $d_1 s_2$. As such, $\text{degen}(H_1 \times K_{s_2, t_2}) = d_1 s_2$. By the same reasoning:

$\text{degen}(H_1 \times K_{1,\Delta_2}) = d_1$, $\text{degen}(K_{s_1,t_1} \times H_2) = s_1 d_2$, and $\text{degen}(K_{1,\Delta_1} \times H_2) = d_2$. Again, the degeneracy is at most $\tilde{d}$ for each of the above graphs. Hence, $\text{degen}(J_1 \times J_2) \leq \tilde{d}$ whenever $J_i \in \{H_i, K_{s_i,t_i}, K_{1,\Delta_i}\}$ for $i \in \{1, 2\}$, as required. $\square$

We conclude that the upper and lower bounds in Theorem 12 are tight.

3.2 | Strong product

We now consider the degeneracy of a strong product. As the next lemma illustrates, more cases arise for this graph product compared to the other two.

Lemma 14. *For all integers $t_i \geq s_i \geq 1$, $\text{degen}(K_{s_1,t_1} \boxtimes K_{s_2,t_2}) = f(s_1, t_1, s_2, t_2)$, where*

$$f(s_1, t_1, s_2, t_2) := \max\{s_1 + s_2 + s_1 s_2, \min\{t_1 + t_2, s_1(t_2 + 1), s_2(t_1 + 1)\}, \min\{s_1 t_2, s_2 t_1\}\}.$$

Proof. Let $\hat{d} := f(s_1, t_1, s_2, t_2)$. Let (S_1, T_1) and (S_2, T_2) be the bipartition of K_{s_1,t_1} and K_{s_2,t_2} , respectively. Let $G := K_{s_1,t_1} \boxtimes K_{s_2,t_2}$. Let $A := S_1 \times S_2$, $B := S_1 \times T_2$, $C := T_1 \times S_2$, and $D := T_1 \times T_2$.

We first prove the lower bound by showing that there exists a subgraph of G with minimum degree $\hat{d}$. Observe that $\delta(G) = s_1 + s_2 + s_1 s_2$. By Observation 11, there exists a subgraph J of $K_{s_1,t_1} \times K_{s_2,t_2} \subseteq G$ with $\delta(J) = \min\{s_1 t_2, s_2 t_1\}$. Let H be the subgraph of G induced by $A \cup B \cup C$. For this subgraph, $\deg_H(a) = t_1 + t_2$, $\deg_H(b) = s_2(t_1 + 1)$, and $\deg_H(c) = s_1(t_2 + 1)$ for every $a \in A$, $b \in B$ and $c \in C$. As such, $\delta(H) = \min\{t_1 + t_2, s_1(t_2 + 1), s_2(t_1 + 1)\}$. Therefore, either G, J or H has minimum degree $\hat{d}$.

We now prove the upper bound. Let Z be a subgraph of G . We proceed by case analysis to show that the minimum degree of Z is at most $\hat{d}$.

First, if $V(Z)$ is a subset of either A, B, C or D , then $\delta(Z) = 0$ since these are independent sets. So we may assume that $V(Z)$ intersects at least two of those sets. Now if there exists a vertex $v \in V(Z) \cap D$, then $\deg_Z(v) \leq s_1 + s_2 + s_1 s_2 \leq \hat{d}$. So we may assume that $V(Z) \cap D = \emptyset$. Now if $V(Z) \cap A$, $V(Z) \cap B$, and $V(Z) \cap C$ are all nonempty, then $\delta(Z) \leq \min\{t_1 + t_2, s_1(t_2 + 1), s_2(t_1 + 1)\}$. Furthermore, if $V(Z) \subseteq B \cup C$, then $\delta(Z) \leq \min\{s_1 t_2, s_2 t_1\}$. Finally, if $V(Z) \subseteq A \cup B$ or $V(Z) \subseteq A \cup C$, then $\delta(Z) \leq \max\{s_1, s_2\}$. In each case, $\delta(Z) \leq \hat{d}$ as required. $\square$

The next theorem provides tight bounds for the degeneracy of a strong product.

Theorem 15. *For $i \in \{1, 2\}$, let G_i be a graph with maximum degree Δ_i and $\text{degen}(G_i) = d_i$ that contains K_{s_i,t_i} as a subgraph where $s_i \leq t_i$. Then*

$$g(d_1, \Delta_1, s_1, t_1, d_2, \Delta_2, s_2, t_2) \leq \text{degen}(G_1 \boxtimes G_2) \leq h(d_1, \Delta_1, d_2, \Delta_2)$$

where f is specified by Lemma 14 and

$$\begin{aligned} g(d_1, \Delta_1, s_1, t_1, d_2, \Delta_2, s_2, t_2) &= \max\{d_1 + d_2 + d_1 d_2, f(s_1, t_1, s_2, t_2), \min\{\Delta_1, \Delta_2\} + 1\}, \\ h(d_1, \Delta_1, d_2, \Delta_2) &= d_1 + d_2 + \min\{d_1 \Delta_2, d_2 \Delta_1\}. \end{aligned}$$

Proof. We first prove the lower bound. Let Q_i be a subgraph of G_i with $\delta(Q_i) = d_i$. Then $\delta(Q_1 \boxtimes Q_2) = d_1 + d_2 + d_1 d_2$ and thus, $\text{degen}(G_1 \boxtimes G_2) \geq d_1 + d_2 + d_1 d_2$. Furthermore, since G_i contains K_{1, Δ_i} and K_{s_i, t_i} as a subgraph, by Lemma 14, $\text{degen}(G_1 \boxtimes G_2) \geq f(s_1, t_1, s_2, t_2)$ and $\text{degen}(G_1 \boxtimes G_2) \geq \min\{\Delta_1, \Delta_2\} + 1$.

We now prove the upper bound. Let Z be a subgraph of $G_1 \boxtimes G_2$. Our goal is to show that $\delta(Z) \leq h(d_1, \Delta_1, d_2, \Delta_2)$. Without loss of generality, $d_1 \Delta_2 \leq d_2 \Delta_1$. Let Z be a subgraph of $G_1 \boxtimes G_2$ and let X be the projection of Z onto G_1 . Since G_1 is d_1 -degenerate, there exists a vertex $v_1 \in V(X)$ with $\deg_X(v_1) \leq d_1$. Let Y be the subgraph of G_2 induced by the set of vertices v_2 in G_2 such that $(v_1, v_2) \in V(Z)$. Since G_2 is d_2 -degenerate, there exists a vertex $v_2 \in V(Y)$ with $\deg_Y(v_2) \leq d_2$. By construction of Y , $(v_1, v_2) \in V(Z)$. By the definition of the strong product,

$$\deg_Z((v_1, v_2)) \leq |N_X(v_1)| + |N_Y(v_2)| + |N_X(v_1)| |N_Y(v_2)| \leq d_1 + d_2 + d_1 \Delta_2.$$

Hence $\delta(Z) \leq h(d_1, \Delta_1, d_2, \Delta_2)$ as required. $\square$

We now construct a family of graphs that realises the lower bound in Theorem 15.

Lemma 16. *For all integers $d_i, \Delta_i, s_i, t_i \geq 1$, where $s_i \leq d_i \leq \Delta_i$ and $s_i \leq t_i \leq \Delta_i$ for $i \in \{1, 2\}$, there exists graphs G_1 and G_2 , where for $i \in \{1, 2\}$, G_i has maximum degree Δ_i , $\text{degen}(G_i) = d_i$, and G_i contains K_{s_i, t_i} as a subgraph, such that*

$$\text{degen}(G_1 \boxtimes G_2) = g(d_1, \Delta_1, s_1, t_1, d_2, \Delta_2, s_2, t_2)$$

where g is specified in Theorem 15.

Proof. Let $\tilde{d} := g(d_1, \Delta_1, s_1, t_1, d_2, \Delta_2, s_2, t_2)$. Our proof parallels the proof of Lemma 13. For $i \in \{1, 2\}$, let H_i be a d_i -regular graph and let G_i be the disjoint union of H_i , K_{s_i, t_i} and K_{1, Δ_i} . Then G_i has maximum degree Δ_i , $\text{degen}(G_i) = d_i$, and G_i contains K_{s_i, t_i} and K_{1, Δ_i} as subgraphs.

We now show that $\text{degen}(G_1 \boxtimes G_2) = \tilde{d}$. Theorem 15 provides the lower bound. For the upper bound, our goal is to show that $\text{degen}(J_1 \boxtimes J_2) \leq \tilde{d}$ whenever $J_i \in \{H_i, K_{s_i, t_i}, K_{1, \Delta_i}\}$ for $i \in \{1, 2\}$. We proceed by case analysis.

First, $\text{degen}(H_1 \boxtimes H_2) = d_1 + d_2 + d_1 d_2$ since H_1 and H_2 are regular. By Lemma 14,

$$\text{degen}(K_{s_1, t_1} \boxtimes K_{s_2, t_2}) = f(s_1, t_1, s_2, t_2) \quad \text{where } f \text{ is specified by Lemma 14,}$$

$$\text{degen}(K_{s_1, t_1} \boxtimes K_{1, \Delta_2}) = \max\{2s_1 + 1, \min\{t_1, s_1 \Delta_2\}\},$$

$$\text{degen}(K_{1, \Delta_1} \boxtimes K_{s_2, t_2}) = \max\{2s_2 + 1, \min\{s_2 \Delta_1, t_2\}\}, \text{ and}$$

$$\text{degen}(K_{1, \Delta_1} \boxtimes K_{1, \Delta_2}) = \min\{\Delta_1, \Delta_2\} + 1.$$

For each of the above graphs, the degeneracy is at most $\tilde{d}$. Now consider the graph $H_1 \boxtimes K_{s_2, t_2}$. Each vertex in $H_1 \boxtimes K_{s_2, t_2}$ has degree $d_1 + s_2 + d_1 s_2$ or $d_1 + t_2 + d_1 t_2$. Let $A := \{v \in V(H_1 \boxtimes K_{s_2, t_2}) : \deg(v) = d_1 + s_2 + d_1 s_2\}$ and $B := \{v \in V(H_1 \boxtimes K_{s_2, t_2}) : \deg(v) = d_1 + t_2 + d_1 t_2\}$. Let Z be a subgraph of $H_1 \boxtimes K_{s_2, t_2}$. If $V(Z) \cap A$ is nonempty, then $\delta(Z) \leq d_1 + s_2 + d_1 s_2$. Otherwise, $V(Z) \subseteq B$ in which case $\delta(Z) \leq d_1$. Thus, $\text{degen}(H_1 \boxtimes K_{s_2, t_2}) = d_1 + s_2 + d_1 s_2$. By the same reasoning: $\text{degen}(H_1 \boxtimes K_{1, \Delta_2}) =$

$2d_1 + 1$, $\text{degen}(K_{s_1, t_1} \boxtimes H_2) = d_2 + s_1 + d_2 s_1$, and $\text{degen}(K_{1, \Delta_1} \boxtimes H_2) = 2d_2 + 1$. Again, the degeneracy is at most $\tilde{d}$ for each of the above graphs. Having considered all possibilities, it follows that $\text{degen}(J_1 \boxtimes J_2) \leq \tilde{d}$ whenever $J_i \in \{H_i, K_{s_i, t_i}, K_{1, \Delta_i}\}$ for $i \in \{1, 2\}$, as required. $\square$

The next lemma describes a family of graphs that realises the upper bound in Theorem 15.

Lemma 17. *For all integers $k_1, k_2, d_1, d_2 \geq 1$ there exists graphs G_1 and G_2 , where for $i \in \{1, 2\}$, G_i has maximum degree $\Delta_i := k_i d_i$, and $\text{degen}(G_i) = d_i$, such that*

$$\text{degen}(G_1 \boxtimes G_2) = d_1 + d_2 + \min\{d_1 \Delta_2, d_2 \Delta_1\}.$$

Proof. For $i \in \{1, 2\}$, let G_i be a graph with vertex partition (A_i, B_i) , where A_i is a clique on $d_i + 1$ vertices and B_i is an independent set of size $(d_i + 1)(k_i - 1)$ such that each $a_i \in A_i$ has $d_i(k_i - 1)$ neighbours in B_i and each $b_i \in B_i$ has d_i neighbours in A_i . Then G_i has maximum degree Δ_i and $\text{degen}(G_i) = d_i$.

We now show that $\text{degen}(G_1 \boxtimes G_2) = d_1 + d_2 + \min\{d_1 \Delta_2, d_2 \Delta_1\}$. Theorem 15 provides the upper bound. Let $X := A_1 \times A_2$, $Y := B_1 \times A_2$, and $Z := A_1 \times B_2$. Let H be the subgraph of $G_1 \boxtimes G_2$ induced on $X \cup Y \cup Z$. Let $x \in X, y \in Y$ and $z \in Z$. Then

$$\begin{aligned} \deg_H(x) &= |N_H(x) \cap X| + |N_H(x) \cap Y| + |N_H(x) \cap Z| \\ &= ((d_1 + 1)(d_2 + 1) - 1) + (d_2(\Delta_1 - d_1)) + (d_1(\Delta_2 - d_2)), \\ \deg_H(y) &= |N_H(y) \cap X| + |N_H(y) \cap Y| + |N_H(y) \cap Z| \\ &= (d_1(d_2 + 1)) + (d_2) + (d_1(\Delta_2 - d_2)) = d_1 + d_2 + \Delta_2 d_1, \text{ and} \\ \deg_H(z) &= |N_H(z) \cap X| + |N_H(z) \cap Y| + |N_H(z) \cap Z| \\ &= (d_2(d_1 + 1)) + (d_2(\Delta_1 - d_1)) + (d_1) = d_1 + d_2 + d_2 \Delta_1. \end{aligned}$$

In which case, the minimum degree of H is $d_1 + d_2 + \min\{d_1 \Delta_2, d_2 \Delta_1\}$ as required. $\square$

We conclude that the upper and lower bounds in Theorem 15 are tight.

4 | PATHWIDTH AND TREewidth

We now consider the treewidth and pathwidth of a graph product. For graphs G and H , an H -decomposition of G consists of a collection $\mathcal{B} = \{B_x \subseteq V(G) : x \in V(H)\}$ of bags with the following properties.

- For each $v \in V(G)$, $H[\{x \in V(H) : v \in B_x\}]$ is a nonempty connected subgraph of H ; and
- For each $uv \in E(G)$, there is a node $x \in V(H)$ such that $u, v \in B_x$.

The *width* of an H -decomposition is $\max\{|B_x| - 1 : x \in V(H)\}$. We denote an H -decomposition by the pair $(H, \mathcal{B})$. A *tree-decomposition* is a T -decomposition for some tree T . The *treewidth* of a graph G , $\text{tw}(G)$, is the minimum width of a tree-decomposition of G . A *path-decomposition* is a P -decomposition for some path P . The *pathwidth* of a graph G , $\text{pw}(G)$, is the

minimum width of a path-decomposition of G . Since a path is a tree, $\text{tw}(G) \leq \text{pw}(G)$. Treewidth and pathwidth, respectively, measure how similar a graph is to a tree and to a path. These two graph parameters are minor-closed. Tree-decompositions and path-decompositions were introduced by Robertson and Seymour [61, 62]; the more general notion of H -decompositions were introduced by Diestel and Kühn [18].

Let G be a graph. We say that $X, Y \subseteq V(G)$ *touch* if $X \cap Y \neq \emptyset$ or there is an edge of G between X and Y . A *bramble*, $\mathcal{B}$, is a set of pairwise touching connected subgraphs. A set $S \subseteq V(G)$ is a *hitting set* of $\mathcal{B}$ if S intersects every element of $\mathcal{B}$. The *order* of $\mathcal{B}$ is the minimum size of a hitting set of $\mathcal{B}$. The canonical example of a bramble of order ℓ is the set of crosses (union of a row and column) in the $\ell \times \ell$ grid. The following ‘Treewidth Duality Theorem’ shows the intimate relationship between treewidth and brambles.

Theorem 18 (Seymour and Thomas [65]). *A graph G has treewidth at least ℓ if and only if G contains a bramble of order at least $\ell + 1$.*

Many papers have studied the treewidth and pathwidth for various families of graphs that are defined by a graph product, see [14, 16, 20, 32, 35–37, 46, 50, 52, 55, 56]. In this section, we continue this work by presenting new lower bounds for cartesian and strong product as well as characterising when the cartesian, direct and strong products have bounded treewidth and when they have bounded pathwidth.

4.1 | Cartesian and strong product

First, consider the treewidth of the cartesian and strong products. The following upper bound is well-known (see [6] for an implicit proof).

Lemma 19. *For all graphs G_1 and G_2 ,*

$$\begin{aligned} \text{tw}(G_1 \square G_2) &\leq \text{tw}(G_1 \boxtimes G_2) \leq (\text{tw}(G_1) + 1)v(G_2) - 1, \text{ and} \\ \text{pw}(G_1 \square G_2) &\leq \text{pw}(G_1 \boxtimes G_2) \leq (\text{pw}(G_1) + 1)v(G_2) - 1. \end{aligned}$$

The proof of Lemma 19 in [6] shows the following, more general result which we use in Section 4.3.

Lemma 20. *For all graphs G_1, G_2 and H , if G_1 has an H -decomposition with width at most k , then $G_1 \boxtimes G_2$ has an H -decomposition with width at most $(k + 1)v(G_2) - 1$.*

Proof Sketch. Let $(H, \mathcal{W})$ be an H -decomposition of G_1 with width k . Modify $(H, \mathcal{W})$ to obtain a tree-decomposition $(H, \mathcal{B})$ of $G_1 \boxtimes G_2$ by setting $B_t := \{(v, u) : v \in W_t, u \in V(G_2)\}$ for all $t \in V(H)$. Then $(H, \mathcal{B})$ is an H -decomposition of $G_1 \boxtimes G_2$ with width at most $(k + 1)v(G_2) - 1$. $\square$

A natural question is whether the upper bound in Lemma 19 is tight up to a constant factor. The following result shows that this is not the case.

Proposition 21. For all $n \geq k + 1 \geq 0$, there exists a connected graph $G_{k,n}$ such that $\text{tw}(G_{k,n}) = k$ and $v(G_{k,n}) = n$ and

$$\text{tw}(G_{k,n} \boxtimes G_{k,n}) = \Theta(n + k^2).$$

Proof. We make no attempt to optimise the constants in this proof. Let $G_{k,n}$ be the graph that consists of a path $P_{\tilde{n}} = (v_0, \dots, v_{\tilde{n}-1})$ on $\tilde{n} = n - k$ vertices and a complete graph K_{k+1} where $V(P_{\tilde{n}}) \cap V(K_k) = \{v_0\}$. Then $\text{tw}(G_{k,n}) = k$ and $v(G_{k,n}) = n$.

We now show that $\text{tw}(G_{k,n} \boxtimes G_{k,n}) = \Theta(n + k^2)$. For the lower bound, since $G_{k,n}$ is connected, by Theorem 22, $\text{tw}(G_{k,n} \boxtimes G_{k,n}) \geq n - 1$. Moreover, since $K_k \boxtimes K_k \simeq K_{k^2}$, $\text{tw}(G_{k,n} \boxtimes G_{k,n}) \geq \text{tw}(K_{k^2}) = k^2 - 1$. Thus $\text{tw}(G_{k,n} \boxtimes G_{k,n}) = \Omega(n + k^2)$.

For the upper bound, we construct a tree decomposition of $G_{k,n} \boxtimes G_{k,n}$ with width $O(n + k^2)$. We begin by specifying the bags. For $i, j, \ell \in [\tilde{n} - 1]$, let

$$\begin{aligned} X &:= \{(v_0, u), (u, v_0) : u \in V(G_{k,n})\}, \\ W_y &:= \{(a, b) : a, b \in K_k\} \cup X, \\ C_{u_i} &:= \{(a, v_{j-1}), (a, v_j) : a \in K_k\} \cup X, \\ D_{w_j} &:= \{(v_{i-1}, a), (v_i, a) : a \in K_k\} \cup X, \text{ and} \\ L_{z_\ell} &:= \{(v_{\ell-1}, v_s), (v_\ell, v_s), (v_s, v_{\ell-1}), (v_s, v_\ell) : s \in [\ell, n - 1]\} \cup X. \end{aligned}$$

Observe that $|X| \leq 2n$, $|W_y| \leq 2n + k^2$, $|C_i| = |D_i| \leq 2n + 2k$ and $|L_{w_\ell}| \leq 6n$. Moreover, observe that $V(K_k \boxtimes K_k) \subseteq W_y$, $V(K_k \boxtimes P_{\tilde{n}}) \subseteq \bigcup_{i \in [\tilde{n}-1]} C_{u_i}$, $V(P_{\tilde{n}} \boxtimes K_k) \subseteq \bigcup_{i \in [\tilde{n}-1]} D_{w_i}$, and $V(P_{\tilde{n}} \boxtimes P_{\tilde{n}}) \subseteq \bigcup_{\ell \in [\tilde{n}-1]} L_{z_\ell}$. Thus, every bag has size $O(n + k^2)$ and every vertex is in a bag. For the tree to index the decomposition, let $P^{(C)} := (u_1, \dots, u_{\tilde{n}-1})$, $P^{(D)} := (w_1, \dots, w_{\tilde{n}-1})$, and $P^{(L)} := (z_1, \dots, z_{\tilde{n}-1})$ be paths on $\tilde{n} - 1$ vertices. Let T be the tree obtained by taking the disjoint union of $P^{(C)}$, $P^{(D)}$ and $P^{(L)}$, then adding the vertex y and the edges yu_1, yw_1, yz_1 (see

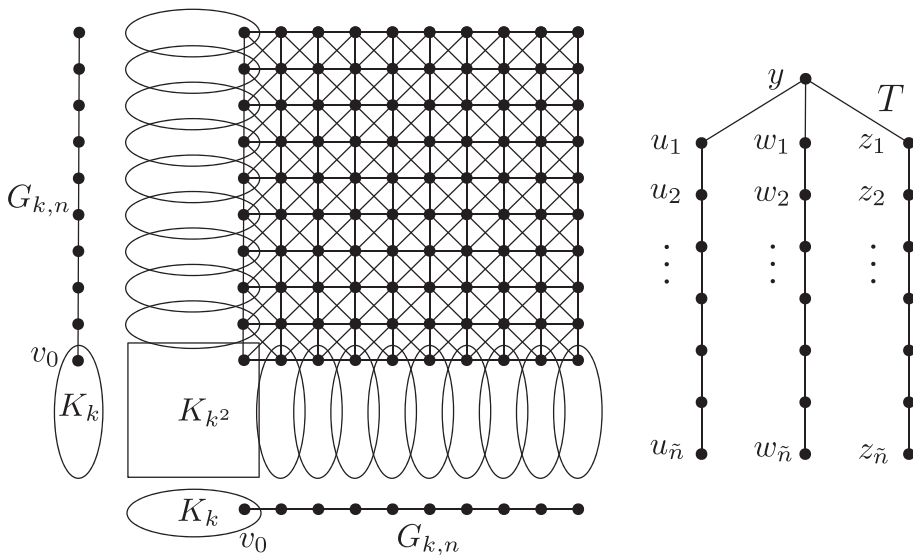


FIGURE 1 $G_{k,n} \boxtimes G_{k,n}$ with the tree T .

Figure 1). Let $\mathcal{W} := \{W_t : t \in V(T)\}$. We claim that $(T, \mathcal{W})$ is a tree decomposition of $G_{k,n} \boxtimes G_{k,n}$.

As noted earlier, every vertex is in a bag. Let $(a_1, a_2)(b_1, b_2) \in E(G_{k,n} \boxtimes G_{k,n})$. Suppose that a_1 or a_2 is equal to v_0 . Then $(a_1, a_2) \in X$. Thus, since $(b_1, b_2) \in W_{t'}$ for some $t' \in V(T)$ and $X \subseteq W_t$ for all $t \in V(T)$, we have $(a_1, a_2)(b_1, b_2) \in W_{t'}$. Now suppose that $a_1, a_2 \in K_{k+1} \setminus \{v_0\}$. Then $N((a_1, a_2)) \subseteq K_{k+1} \boxtimes K_{k+1}$ in which case $(a_1, a_2), (b_1, b_2) \in W_y$. Now if $a_1 \in K_{k+1} \setminus \{v_0\}$ and $a_2 = v_i$ for some $i \in [\tilde{n} - 1]$, then $N((a_1, a_2)) \subseteq C_{u_{i-1}} \cup C_{u_i}$ in which case $(a_1, a_2), (b_1, b_2) \in W_{u_{i'}}$ for $i' \in \{i - 1, i\}$. Similarly, if $a_2 \in K_{k+1} \setminus \{v_0\}$ and $a_1 = v_i$ for some $j \in [\tilde{n} - 1]$, then $(a_1, a_2), (b_1, b_2) \in W_{w_{j'}}$ for $j' \in \{j - 1, j\}$. Finally, if $a_1 = v_i$ and $a_2 = v_j$ for some $i, j \in [\tilde{n} - 1]$, then $N(a_1, a_2) \subseteq L_{w_{\ell-1}} \cup L_{w_\ell}$ and thus $(a_1, a_2), (b_1, b_2) \in W_{u_{\ell'}}$ for some $\ell' \in [\ell - 1, \ell]$ where $\ell := \min\{i, j\}$. As such, every edge is in a bag.

It remains to show that for any $(a_1, a_2) \in V(G_{k,n} \boxtimes G_{k,n})$, the subtree $T^{(a_1, a_2)} := T[t \in V(T) : (a_1, a_2) \in W_t]$ is connected. If $(a_1, a_2) \in X$, then $T^{(a_1, a_2)} = V(T)$. If $a_1, a_2 \in V(K_k) \setminus \{v_0\}$, then $V(T^{(a_1, a_2)}) = \{y\}$. If $a_1 \in V(K_k) \setminus \{v_0\}$ and $a_2 = v_i$ for some $i \in [\tilde{n} - 1]$, then $V(T^{(a_1, a_2)}) = \{u_{i-1}, u_i\}$. If $a_1 = v_j$ for some $j \in [\tilde{n} - 1]$ and $a_2 \in V(K_k) \setminus \{v_0\}$, then $V(T^{(a_1, a_2)}) = \{w_{j-1}, w_j\}$. If $a_1 = v_j$ and $a_2 = v_i$ for some $i, j \in [\tilde{n} - 1]$ then $V(T^{(a_1, a_2)}) = \{z_{\ell-1}, z_\ell\}$, where $\ell := \min\{i, j\}$. In each case, $T^{(a_1, a_2)}$ is connected. Therefore, $(T, \mathcal{W})$ is indeed a tree decomposition of $G_{k,n} \boxtimes G_{k,n}$ with width $O(n + k^2)$ as required. $\square$

We now consider lower bounds for the treewidth of cartesian and strong products. Wood [77] established the following lower bound for highly connected graphs.

Theorem 22 (Wood [77]). *For all k -connected graphs G_1 and G_2 each with at least n vertices,*

$$\text{tw}(G_1 \boxtimes G_2) \geq \text{tw}(G_1 \square G_2) \geq k(n - 2k + 2) - 1.$$

We now present two new lower bounds. For a graph G and $\epsilon \in [\frac{2}{3}, 1)$, a partition (A, S, B) of $V(G)$ is an ϵ -separation if $1 \leq |A|, |B| \leq \epsilon|V(G)|$ and there is no edge between A and B . The order of (A, S, B) is $|S|$. Robertson and Seymour [62] showed that graphs with small treewidth have separations with small order.

Lemma 23 (Robertson and Seymour [62]). *For every $\epsilon \in [\frac{2}{3}, 1)$, every graph with treewidth k has an ϵ -separation of order at most $k + 1$.*

Lemma 24. *For all $\epsilon, \beta \in [\frac{2}{3}, 1)$ where $\beta > \epsilon$ and integers $k, n, m \geq 1$ where $m \geq kn$ and $n \geq \frac{1}{1-\beta}$ and for all connected graphs G and H with m and n vertices respectively such that every β -separation of G has order at least k , every ϵ -separation of $G \square H$ has order at least $(1 - \frac{\epsilon}{\beta})kn$.*

Proof. Let $V(H) := [n]$. Let (A, S, B) be a ϵ -separation of $G \square H$. Our goal is to show that $|S| \geq \left(1 - \frac{\epsilon}{\beta}\right)kn$. For $i \in [n]$, let $G^{(i)}$ be the copy of G in $G \square H$ induced by $\{(v, i) : v \in V(G)\}$. We say that $G^{(i)}$ belongs to A if $|A \cap V(G^{(i)})| \geq \beta m$, and $G^{(i)}$ belongs to B if $|B \cap V(G^{(i)})| \geq \beta m$.

Suppose that some copy $G^{(i)}$ belongs to A and some copy $G^{(j)}$ belongs to B . Let $X := \{v \in V(G) : (v, i) \in A, (v, j) \in B\}$. Thus $|X| \geq (2\beta - 1)m > 0$, which implies $i \neq j$. Since H is connected, for each $x \in X$ there is a path from (x, i) to (x, j) contained within the subgraph of $G \square H$ induced by $\{(x, \ell) : \ell \in V(H)\}$. Since these paths are pairwise disjoint and each path contains a vertex from S , we have $|S| \geq |X| \geq (2\beta - 1)kn \geq \left(1 - \frac{\epsilon}{\beta}\right)kn$.

Now assume, without loss of generality, that no copy of G belongs to B . Say t copies of G belong to A . Then $\beta tm \leq |A| \leq \epsilon nm$, implying that $t \leq \frac{\epsilon n}{\beta}$. Thus, at least $\left(1 - \frac{\epsilon}{\beta}\right)n$ copies of G belong to neither A nor B . Now consider such a copy $G^{(i)}$. If $G^{(i)}$ is contained in $S \cup A$, then $|S \cap V(G^{(i)})| \geq (1 - \beta)m \geq k$. Similarly, if $G^{(i)}$ is contained in $S \cup B$, then $|S \cap V(G^{(i)})| \geq k$. Otherwise, $G^{(i)}$ contains vertices in both A and B , in which case $(A \cap V(G^{(i)}), S \cap V(G^{(i)}), B \cap V(G^{(i)}))$ is a β -separation of $G^{(i)}$ and thus, $|S \cap V(G^{(i)})| \geq k$. Therefore,

$$|S| = \sum_{i=1}^n |V(G^{(i)}) \cap S| \geq \left(1 - \frac{\epsilon}{\beta}\right)kn$$

as required. $\square$

Lemmas 23 and 24 imply the following.

Theorem 25. For all $\epsilon, \beta \in \left[\frac{2}{3}, 1\right)$ where $\beta > \epsilon$ and integers $k, n, m \geq 1$ where $m \geq kn$ and $n \geq \frac{1}{1-\beta}$ and for all connected graphs G and H with m and n vertices respectively such that every β -separation of G has order at least k ,

$$\text{tw}(G \square H) \geq \left(1 - \frac{\epsilon}{\beta}\right)kn - 1.$$

By applying Lemmas 24 and 19, we determine the treewidth of d -dimensional grid graphs up to a constant factor.

Corollary 26. For fixed $d \geq 2$ and all integers $n_1 \geq \dots \geq n_d \geq 1$,

$$\text{tw}(P_{n_1} \square \dots \square P_{n_d}) = \Theta\left(\prod_{j=2}^d n_j\right) \quad \text{and} \quad \text{tw}(P_{n_1} \boxtimes \dots \boxtimes P_{n_d}) = \Theta\left(\prod_{j=2}^d n_j\right).$$

Proof Sketch. We proceed by induction on $d \geq 2$. For the upper bound, apply Lemma 19 by setting $G_1 := P_{n_1} \boxtimes \dots \boxtimes P_{n_{d-1}}$ and $G_2 := P_{n_d}$ with the induction hypothesis that $\text{tw}(P_{n_1} \boxtimes \dots \boxtimes P_{n_d}) \leq \prod_{j=2}^d n_j$ for all $d \geq 2$. For the lower bound, apply Lemma 24 by setting $G := P_{n_1} \square \dots \square P_{n_{d-1}}$ and $H := P_{n_d}$ with the induction hypothesis that for all $d \geq 2$

and $\epsilon \in \left[\frac{2}{3}, 1\right)$, there is a constant $c(d, \epsilon)$ such that for all sufficiently large n_d (as a function of d and ϵ), every ϵ -separation in $P_{n_1} \square \cdots \square P_{n_d}$ has order at least $c(d, \epsilon) \prod_{j=2}^d n_j$. The lower bound then follows by Lemma 23. $\square$

This result demonstrates that d -dimensional grids are a family of graphs for which Theorem 25 is tight up to a constant factor, whereas the lower bound given by Theorem 22 is not of the correct order for this family.

We now present another lower bound in terms of the Hadwiger number. For a graph G , the *Hadwiger number* of G , $\eta(G)$, is the maximum integer t such that K_t is a minor of G .

Theorem 27. *For all graphs G and H ,*

$$\text{tw}(G \boxtimes H) \geq \eta(H)(\text{tw}(G) + 1) - 1.$$

Proof. Let $(B_i)_{i \in V(K_t)}$ be a model of K_t in H where $t := \eta(H)$. By Theorem 18, there is a bramble $\mathcal{B}$ in G of such that every hitting set of $\mathcal{B}$ has order at least $\text{tw}(G) + 1$. For $X \in \mathcal{B}$ and $i \in [t]$, let $(X, i) := \{(v, u) \in V(G \boxtimes H) : v \in X, u \in B_i\}$. Let $G_{X,i}$ denote the subgraph of $G \boxtimes H$ that is induced by (X, i) . Since $G[X]$ and $H[B_i]$ are connected, $G_{X,i}$ is connected. For $i \in [t]$, let $\mathcal{B}_i := \{(X, B_i) : X \in \mathcal{B}\}$. Then $\mathcal{B}_i$ is a bramble for $G \boxtimes (H[B_i])$.

Let $\mathcal{C} := \{(X, B_i) : X \in \mathcal{B}, i \in [t]\}$. Consider $(X, B_i), (Y, B_j) \in \mathcal{C}$. Since X and Y touch in G , for some vertices $v \in X$ and $w \in Y$, either $v = w$ or $vw \in E(G)$. Moreover, there exists $u_i \in B_i$ and $u_j \in B_j$ such that $u_i = u_j$ (if $i = j$) or $u_i u_j \in E(H)$. Thus, in $G \boxtimes H$, the vertices (v, u_i) and (w, u_j) are adjacent or equal. Since $(v, u_i) \in (X, B_i)$ and $(w, u_j) \in (Y, B_j)$, the sets (X, B_i) and (Y, B_j) touch. Hence $\mathcal{C}$ is a bramble in $G \boxtimes H$. Let J be a hitting set for $\mathcal{C}$. For each $i \in [t]$, let $J_i := \{(v, u) \in J : u \in B_i\}$. Thus J_i is a hitting set for the bramble $\mathcal{B}_i$ (in $G \boxtimes B_i$). By Theorem 18, $|J_i| \geq \text{tw}(G) + 1$. Since the J_i 's are pairwise disjoint, $|J| \geq t(\text{tw}(G) + 1)$. Hence $\text{tw}(G \boxtimes H) \geq t(\text{tw}(G) + 1) - 1$ by Theorem 18. $\square$

Theorem 27 and Lemma 19 together imply that for every graph G ,

$$\text{tw}(G \boxtimes K_n) = (\text{tw}(G) + 1)n - 1.$$

The next two theorems characterise when the cartesian and strong products have bounded treewidth and when they have bounded pathwidth.

Theorem 28. *The following are equivalent for monotone graph classes $\mathcal{G}_1$ and $\mathcal{G}_2$:*

1. $\mathcal{G}_1 \boxtimes \mathcal{G}_2$ has bounded treewidth;
2. $\mathcal{G}_1 \square \mathcal{G}_2$ has bounded treewidth;
3. both $\mathcal{G}_1$ and $\mathcal{G}_2$ have bounded treewidth and $\tilde{v}(\mathcal{G}_1)$ or $\tilde{v}(\mathcal{G}_2)$ is bounded; and
4. $\mathcal{G}_1 \boxtimes \mathcal{G}_2$ has bounded Hadwiger number;

Proof. Building on the work of Wood [76], Pecaninovic [57] showed that (3) and (4) are equivalent. Since treewidth is closed under subgraphs, (1) implies (2). By Lemma 19, (3) implies (1). To show that (2) implies (3), suppose that $\mathcal{G}_1 \square \mathcal{G}_2$ has bounded treewidth. If

either $\mathcal{G}_1$ or $\mathcal{G}_2$ have unbounded treewidth, then $\mathcal{G}_1 \square \mathcal{G}_2$ has unbounded treewidth, a contradiction. If the connected graphs in both $\mathcal{G}_1$ and $\mathcal{G}_2$ have unbounded order, then by Theorem 22, $\mathcal{G}_1 \square \mathcal{G}_2$ has unbounded treewidth, a contradiction. As such, both $\mathcal{G}_1$ and $\mathcal{G}_2$ have bounded treewidth and $\tilde{v}(\mathcal{G}_1)$ or $\tilde{v}(\mathcal{G}_2)$ is bounded. $\square$

We omit the proof for the following theorem as it is identical to Theorem 28.

Theorem 29. *The following are equivalent for monotone graph classes $\mathcal{G}_1$ and $\mathcal{G}_2$:*

1. $\mathcal{G}_1 \boxtimes \mathcal{G}_2$ has bounded pathwidth;
2. $\mathcal{G}_1 \square \mathcal{G}_2$ has bounded pathwidth; and
3. both $\mathcal{G}_1$ and $\mathcal{G}_2$ have bounded pathwidth and $\tilde{v}(\mathcal{G}_1)$ or $\tilde{v}(\mathcal{G}_2)$ is bounded.

We conclude this section with some open problems. First and foremost, can we determine the treewidth of $G \boxtimes H$ up to a constant factor? A first step in resolving this question is to determine if Theorem 27 can be strengthened by showing that there exists a constant $c > 0$ such that $\text{tw}(G \boxtimes H) \geq c \text{tw}(G) \text{tw}(H)$ for all graphs G and H . Similar questions arise for the cartesian product. It is also open whether there exist a constant $c > 0$ such that $\text{tw}(G \boxtimes H) \leq c \text{tw}(G \square H)$ for all graphs G and H .

4.2 | Direct product with K_2

We now consider when a direct product has bounded treewidth and when it has bounded pathwidth. In comparison with the other two products, characterisation for this product is more involved. Before considering the direct product of two classes of graphs, we first investigate the direct product of a class of graphs with a single graph. Lemma 19 immediately provides an upper bound for the treewidth and pathwidth of the direct product of a class of graphs with a single graph. The challenge therefore lies in proving a lower bound. In related results, Bottreau and Métivier [10] demonstrated that for every graph G and nonbipartite graph H , G is a minor of $G \times H$, and hence $\text{tw}(G \times H) \geq \text{tw}(G)$ and $\text{pw}(G \times H) \geq \text{pw}(G)$. This lower bound for treewidth was independently shown by Eppstein et al. [28]. It remains to consider the case when H is bipartite. A case of particular interest is when $H = K_2$. Thomassen [70] showed the following.

Theorem 30 (Thomassen [70]). *There exists a function f such that every graph G with $\text{tw}(G) \geq f(k)$ contains a bipartite subgraph $\hat{G} \subseteq G$ such that $\text{tw}(\hat{G}) \geq k$.*

Note that the function f in Theorem 30 is exponential with respect to k . Since every bipartite subgraph of G is also a subgraph of $G \times K_2$, we have the following.

Corollary 31. *For every graph class $\mathcal{G}$ with unbounded treewidth, $\mathcal{G} \times K_2$ also has unbounded treewidth.*

Together with Lemma 19, Corollary 31 demonstrates that $\text{tw}(G)$ and $\text{tw}(G \times K_2)$ are tied. Similarly, for pathwidth, as explained in our companion paper [41], an analogous result to Theorem 30 is implied by the excluded forest minor theorem [3, 61].

Theorem 32 (Hickingbotham and Wood [41]). *There exists a function f such that every graph G with $\text{pw}(G) \geq f(k)$ contains a bipartite subgraph $\hat{G} \subseteq G$ such that $\text{pw}(\hat{G}) \geq k$.*

Corollary 33. *For every graph class $\mathcal{G}$ with unbounded pathwidth, $\mathcal{G} \times K_2$ also has unbounded pathwidth.*

Together with Lemma 19, it follows that $\text{pw}(G)$ and $\text{pw}(G \times K_2)$ are tied. Note that the function f in Theorem 32 is exponential with respect to k . We now strengthen these results by demonstrating that $\text{tw}(G)$ and $\text{tw}(G \times K_2)$ are polynomially tied and then showing that this implies that $\text{pw}(G)$ and $\text{pw}(G \times K_2)$ are polynomially tied.

Theorem 34. *There exists a positive constant c such that $\text{tw}(G \times K_2) \geq k$ for every graph G with $\text{tw}(G) \geq ck^4 \log^{5/2}(k)$.*

To prove Theorem 34, we make use of grid-like-minors which were introduced by Reed and Wood [60]. A *grid-like-minor of order ℓ* in a graph G is a set $\mathcal{P}$ of paths in G such that the intersection graph of $\mathcal{P}$ is bipartite and contains a K_ℓ minor.* Reed and Wood [60] showed the following.

Theorem 35 (Reed and Wood [60]). *For some positive constant c , every graph with treewidth at least $c\ell^4 \sqrt{\log(\ell)}$ contains a grid-like-minor of order ℓ .*

We now explain how to adapt the proof for Theorem 35 to show that if a graph G has sufficiently large treewidth, then $G \times K_2$ contains a grid-like-minor of large order. A key lemma that Reed and Wood used to prove Theorem 35 is the following.

Lemma 36 (Reed and Wood [60]). *For all integers $k, \ell \geq 1$, every graph G with treewidth at least $k\ell - 1$ contains ℓ disjoint paths $P_1, \dots, P_\ell$, and for distinct $i, j \in [\ell]$, G contains k disjoint paths between P_i and P_j .*

Given a black-white colouring of the vertices of a bipartite graph, to *switch* the colouring means to recolour the black vertices white and the white vertices black. For a graph G with vertex-disjoint subgraphs H_1 and H_2 , a path $P = (v_1, \dots, v_n)$ in G *joins* H_1 and H_2 if $V(P) \cap V(H_1) = \{v_1\}$ and $V(P) \cap V(H_2) = \{v_n\}$. We make use of the following lemma from our companion paper [41] which generalises the result by Erdős [30] which states that every graph contains a bipartite subgraph with at least half of the edges.

Lemma 37 (Hickingbotham and Wood [41]). *Let G be a graph and let $H_1, \dots, H_t$ be vertex-disjoint bipartite subgraphs in G . Suppose there exists a set of internally disjoint paths $\mathcal{P} = \{P_1, \dots, P_k\}$ in G such that for all $i \in [k]$, P_i joins H_j and H_ℓ for some distinct $j, \ell \in [t]$ and that P_i and H_b are disjoint for all $b \in [t] \setminus \{j, \ell\}$. Then there exists a subset $S \subseteq \mathcal{P}$ where $|S| \geq k/2$ such that the graph $\hat{G} := (\bigcup_{i \in [t]} H_i) \cup (\bigcup_{S \in S} S)$ is a bipartite subgraph of G .*

*The intersection graph of a set X , whose elements are sets, has vertex set X where distinct vertices are adjacent whenever the corresponding sets have a nonempty intersection.

The next lemma is the main original contribution for this section in which we adapt Lemma 36 to the setting of a direct product with K_2 .

Lemma 38. *Let G be a graph with treewidth at least $2k\ell - 1$ for some integers $k, \ell \geq 1$. Then there exists a set $X \subseteq E(K_\ell)$ where $|X| \geq |E(K_\ell)|/2$ such that $G \times K_2$ contains ℓ -disjoint paths $P_1, \dots, P_\ell$ and for all $ij \in X$, $G \times K_2$ contains k -disjoint paths between P_i and P_j .*

Proof. By Lemma 36, G contains ℓ disjoint paths $\tilde{P}_1, \dots, \tilde{P}_\ell$, and a set $\mathcal{Q}_{i,j}$ of $2k$ disjoint paths between $\tilde{P}_i$ and $\tilde{P}_j$, where $i, j \in [\ell]$ are distinct. Take a proper black-white colouring of $\tilde{P}_1, \dots, \tilde{P}_j$. For a path $Q = (v_1^{(i,j)}, \dots, v_n^{(i,j)}) \in \mathcal{Q}_{i,j}$, we say that it is *agreeable* if there exists a black-white colouring of Q such that the colour of v_1 and v_n corresponds with the colour they were previously assign. Otherwise, we say that Q is *disagreeable*. Observe that if we switch the black-white colouring of $\tilde{P}_j$, then every agreeable path becomes disagreeable and every disagreeable path becomes agreeable. If there are more agreeable paths than disagreeable, then let $\mathcal{M}_{i,j}$ be the set of agreeable paths, otherwise let $\mathcal{M}_{i,j}$ be the set of disagreeable path. By the pigeon-hole principle, $|\mathcal{M}_{i,j}| \geq k$. Note that the set $\mathcal{M}_{i,j}$ of paths are *pairwise-agreeable*, in the sense that if one path in $\mathcal{M}_{i,j}$ is agreeable then all the paths in $\mathcal{M}_{i,j}$ are agreeable.

For each distinct $i, j \in [\ell]$, let $R_{i,j} = (v_1^{(i,j)}, \dots, v_{n_{i,j}}^{(i,j)})$ be an arbitrary path in $\mathcal{M}_{i,j}$. We now define an auxiliary graph J as follows. Let J consist of the ℓ -disjoint paths $\tilde{P}_1, \dots, \tilde{P}_\ell$. For each distinct $i, j \in [\ell]$, add a path $\tilde{R}_{i,j}$ from $v_1^{(i,j)} \in V(\tilde{P}_i)$ to $v_{n_{i,j}}^{(i,j)} \in V(\tilde{P}_j)$ of length $n_{i,j}$ that is internally disjoint from all other vertices in J . Let $\mathcal{R} = \{\tilde{R}_{i,j} : i, j \in [\ell], i \neq j\}$. By Lemma 37, there exists a set $\mathcal{S} \subseteq \mathcal{R}$, where $|\mathcal{S}| \geq |\mathcal{R}|/2$ such that $\tilde{J} = (\bigcup_{i \in [\ell]} \tilde{P}_i) \cup \mathcal{S}$ is bipartite. Let $X = \{ij : R_{i,j} \in \mathcal{S}\}$ and note that $|X| \geq |E(K_\ell)|/2$. Now for all $ij \in X$, since the set of paths $\mathcal{M}_{i,j}$ are pairwise agreeable, it follows that for all $Q = (v_1, \dots, v_n) \in \mathcal{M}_{i,j}$, we can add a path $\hat{Q}$ of length $|Q|$ from $v_1 \in V(\tilde{P}_i)$ to $v_n \in V(\tilde{P}_j)$ to the graph $\tilde{J}$ that is internally disjoint to all other paths without compromising $\tilde{J}$ being bipartite. Do this for all paths in $\mathcal{M}_{i,j}$ that has not yet been considered whenever $ij \in X$ and let $\hat{J}$ be the bipartite graph obtained. Let $\phi : V(\hat{J}) \rightarrow [2]$ be a proper 2-colouring of $\hat{J}$.

Now consider $G \times K_2$. For each $i \in [\ell]$, let P_i be the path in $G \times K_2$ induced by $\{(v, \phi(v)) : v \in V(\tilde{P}_i)\}$. Since $\tilde{P}_i$ and $\tilde{P}_j$ are disjoint for distinct $i, j \in [\ell]$, it follows that $P_1, \dots, P_\ell$ are ℓ -disjoint paths in $G \times K_2$. It remains to show that for all distinct $ij \in X$, there exists k disjoint paths from P_i to P_j . Let $Q = (v_1^{(i,j)}, \dots, v_n^{(i,j)}) \in \mathcal{M}_{i,j}$. By construction of $\hat{J}$, for each $v \in V(\hat{Q})$ there exists a corresponding $\hat{v} \in V(\tilde{Q})$. Let Y be the path in $G \times K_2$ that is induced by $\{(v, \phi(\hat{v})) : v \in V(Q)\}$ and let $\mathcal{Y}_{i,j}$ be the set of such paths. Since the set of paths $\mathcal{M}_{i,j}$ are internally-disjoint, it follows that the set of paths $\mathcal{Y}_{i,j}$ are also internally disjoint. Moreover, $|\mathcal{Y}_{i,j}| = |\mathcal{M}_{i,j}| \geq k$ as required. $\square$

Let $d(\ell)$ be the minimum integer such that every graph with no K_ℓ minor is $d(\ell)$ -degenerate. Kostochka [47, 48] and Thomason [68] independently proved that $d(\ell) \in \Theta(\ell \sqrt{\log(\ell)})$.

Theorem 39 (Kostochka and colleagues [48, 68, 69]). *Every graph that does not contain K_ℓ as a minor is $d(\ell)$ -degenerate where $d(\ell) := c\ell \sqrt{\log(\ell)}$ and thus has average degree at most $2c\ell \sqrt{\log(\ell)}$ where c is a positive constant.*

The proof for the following theorem is a simple adaption of the proof of Theorem 35 in [60].

Theorem 40. *For every graph G with treewidth at least $c\ell^4 \log^{5/2}(\ell)$, $G \times K_2$ contains a grid-like-minor of order ℓ for some constant c .*

Proof. Let $t := \lceil 2(2c\ell\sqrt{\log(\ell)} + 1) \rceil$ and $k := \lceil 4e \binom{t}{2} d(t) \rceil$ where c is specified by Theorem 39. Let G be a graph with treewidth at least $c\ell^4 \log^{5/2}(\ell)$ which is at least $2kt - 1$ for an appropriate value of c . By Lemma 38, there exists a set $X \subseteq E(K_t)$ where $|X| \geq |E(K_t)|/2$ such that $G \times K_2$ contains t -disjoint paths $P_1, \dots, P_t$ and for all $ij \in X$, $G \times K_2$ contains a set Q_{ij} of k -disjoint paths between P_i and P_j . Let J be the subgraph of K_t with vertex set $V(K_t)$ and edge set X . By Theorem 39, J contains K_ℓ as a minor. From here on in, the rest of the proof follows [60, theorem 1.2].

For distinct $ij, ab \in X$, let $H_{ij,a,b}$ be the intersection graph of $Q_{ij} \cup Q_{a,b}$. Since $H_{ij,a,b}$ is bipartite, if K_t is a minor of $H_{ij,a,b}$ then we are done. So assume that K_t is not a minor of $H_{ij,a,b}$. By Theorem 39, $H_{ij,a,b}$ is $d(t)$ -degenerate. Let H be the intersection graph of $\cup\{Q_{ij} : ij \in X\}$. Then H is $|X|$ -colourable where each colour class is some Q_{ij} . Since each colour class has k vertices, and each pair of colour-classes in H induce a $d(t)$ -degenerate subgraph, then by [60, lemma 4.3], H has an independent set with one vertex from each colour class. That is, in each set Q_{ij} there is a path Q_{ij} such that $Q_{ij} \cap Q_{a,b} = \emptyset$ for distinct $ij, ab \in X$. Consider the set of paths $\mathcal{P} := \{P_i : i \in [t]\} \cup \{Q_{ij} : ij \in X\}$. The intersection graph of $\mathcal{P}$ is bipartite and contains the 1-subdivision of J . Since J contains K_ℓ as a minor, $\mathcal{P}$ is a grid-like-minor of order ℓ in G . $\square$

Reed and Wood [60] showed that the treewidth of a grid-like-minor of order ℓ is at least $\lfloor \ell/2 \rfloor - 1$. As such, Theorem 40 implies Theorem 34, and $\text{tw}(G)$ and $\text{tw}(G \times K_2)$ are polynomially tied.

We now show that Theorem 34 implies a polynomial lower bound for the pathwidth of $G \times K_2$. Groenland et al. [33] proved the following.

Theorem 41 [33]. *For all integers $t, h \geq 1$, for every graph G with $\text{tw}(G) \geq t - 1$, either $\text{pw}(G) \leq th + 1$ or G contains a subdivision of a complete binary tree of height h .*

It is well-known that the pathwidth of a subdivision of a complete binary tree with height $2k$ is at least k (see [64]). Therefore, Theorems 41 and 34 imply the following.

Theorem 42. *There exists some positive constant c such that $\text{pw}(G \times K_2) \geq k$ for every graph G with $\text{pw}(G) \geq ck^5 \log^{5/2}(k)$.*

By Lemma 19 and Theorem 42, we conclude that $\text{pw}(G)$ and $\text{pw}(G \times K_2)$ are polynomially tied.

4.3 | Direct product

In this section, we characterise when the direct product of two classes of graphs has bounded treewidth (Theorem 51) and when it has bounded pathwidth (Theorem 52). We begin with

some definitions. For every integer $k \geq 0$, the k -daddy-longlegs $W^{(k)}$ is the tree with $V(W^{(k)}) = \{r, u_1, \dots, u_k, v_1, \dots, v_k\}$ and $E(W^{(k)}) = \{ru_i, u_i v_i : i \in [k]\}$; see Figure 2. Let G be a graph. The *daddy-longlegs number* of G , $\text{dll}(G)$, is the maximum integer $k \geq 0$ such that $W^{(k)}$ is a minor of G . The *path number* of G , $\text{path}(G)$, is the maximum integer $n \geq 0$ such that G contains a path on n vertices. Clearly the path number of a graph is at least its diameter plus 1. A set $A \subseteq V(G)$ is a *vertex cover* of G if $G - A$ is an independent set. The *vertex cover number* of G , $\tau(G)$, is the minimum size of a vertex cover of G .

We now work towards showing that graphs with bounded daddy-longlegs number and bounded path-number have bounded vertex cover number. We begin with some basic lemmas.

Lemma 43. *For all integers $j, n \geq 1$, if a tree T has at most j leaves and $\text{path}(T) \leq n$, then $v(T) \leq \lceil \frac{j}{2} \text{path}(T) \rceil$.*

Proof. We proceed by induction on n . For $n = 1$, the claim holds trivially. Now suppose the claim holds up to $n - 1$. Let T be a tree with at most j leaves and $\text{path}(T) \leq n$. Let T_L be the subtree of T obtained by deleting its leaves. Then T_L has at most j leaves. If T contains a path P with n vertices, then the endpoints of P are leaves in T . If T contains a path P with $n - 1$ vertices, then at least one of the endpoints of T is a leaf. As such, $\text{path}(T_L) \leq n - 2$. By induction, $v(T_L) \leq \lceil j(n - 2)/2 \rceil$. Since at most j vertices were deleted from T to obtain T_L , we have $v(T) \leq v(T_L) + j \leq \lceil jn/2 \rceil$ as required. $\square$

The next lemma is important for our upcoming characterisations.

Lemma 44. *For every connected graph G , $\tau(G) \leq \lceil (\text{dll}(G) + 1) \text{path}(G)/2 \rceil$.*

Proof. Let T be a Depth-First-Search spanning tree of G rooted at some $v \in V(G)$. Note that v may be a leaf vertex in T . Let L be the set of leaves of T excluding v and let $T_L := T - L$. Since T is a Depth-First-Search tree, L is an independent set, and hence $V(T_L)$ is a vertex cover of G . We now bound $v(T_L)$. Let $\tilde{L} := \{\tilde{u}_1, \dots, \tilde{u}_j\}$ be the leaves in T_L excluding v . If $j = 0$ then $V(T_L) = \{v\}$ and we are done. Otherwise, for every $i \in [j]$, let $\tilde{v}_i \in L$ where $\tilde{u}_i \tilde{v}_i \in E(T)$. Note that $\tilde{v}_i \neq \tilde{v}_j$ whenever $i \neq j$. Let $T_r := T_L - \tilde{L}$ and let μ be a model of $W^{(j)}$ in G defined by $\mu(r) = T_r$, $\mu(u_i) = \tilde{u}_i$ and $\mu(v_i) = \tilde{v}_i$ for all $i \in [j]$. Then $W^{(j)}$ is a minor of G and hence $j \leq \text{dll}(G)$. Thus T_L has at most $j + 1$ leaves. Since $\text{path}(T_L) \leq \text{path}(G)$, by Lemma 43, $v(T_L) \leq \lceil (\text{dll}(G) + 1) \text{path}(G)/2 \rceil$ as required. $\square$

We now present several lemmas for direct products in the general framework of H -decompositions. The motivation for doing so is that once results are established within this

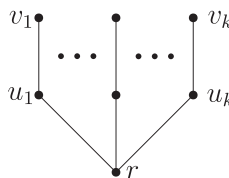


FIGURE 2 The k -daddy-longlegs $W^{(k)}$.

framework, we can quickly deduce bounds for both treewidth and pathwidth. As such, this framework provides a unified approach for which we can tackle both parameters at once.

The *square* of a graph G , denoted G^2 , is the graph with $V(G^2) = V(G)$ where $uv \in E(G^2)$ if $\text{dist}_G(u, v) \leq 2$. For our purposes, the key property of this graph is that $N_G[v]$ is a clique in G^2 for every vertex $v \in V(G)$. The next basic lemma concerning G^2 is useful.

Lemma 45. *For all graphs G and H where G has maximum degree Δ and has an H -decomposition with width at most k , G^2 has an H -decomposition with width at most $(k + 1)(\Delta + 1) - 1$.*

Proof. Let $(H, \mathcal{W})$ be an H -decomposition of G with width at most k . For all $t \in V(H)$, let $B_t := \bigcup_{v \in W_t} N_G[v]$ and let $\mathcal{B} := \{B_t : t \in V(H)\}$. We claim that $(H, \mathcal{B})$ is an H -decomposition of G^2 with width at most $(k + 1)(\Delta + 1) - 1$. Note that for all $v \in V(G^2)$ there exists a node $t_1 \in V(H)$ such that $v \in W_{t_1}$. By construction, $W_{t_1} \subseteq B_{t_1}$ and hence $v \in B_{t_1}$. Hence every vertex in G^2 is in a bag. Let $uv \in E(G^2)$. Now if uv is also an edge in $E(G)$ then there exists a node $t_2 \in V(H)$ such that $\{u, v\} \subseteq W_{t_2}$ and hence $\{u, v\} \subseteq B_{t_2}$. Otherwise, $\text{dist}_G(u, v) = 2$ and hence u and v share a common neighbour $y \in V(G)$ in G . Now since $(H, \mathcal{W})$ is an H -decomposition, there exists a node $t_3 \in V(H)$ such that $y \in W_{t_3}$. So by construction, $\{u, v\} \subseteq N_G[y] \subseteq B_{t_3}$, and hence the endpoints of each edge in G^2 is in a bag.

It remains to show that for all $v \in V(G^2)$, the induced subgraph $H[\{t : v \in B_t\}]$ is connected. For all $v \in V(G^2)$, let $H^{(v)} := H[\{t : v \in B_t\}]$. By construction,

$$H[\{t : v \in B_t\}] = H[\{t : N[v] \cap W_t \neq \emptyset\}] = H\left[\bigcup_{u \in N[v]} \{t : u \in W_t\}\right] = \bigcup_{u \in N[v]} H^{(u)}.$$

Now for all $u \in N_G(v)$, there exists $x \in V(H)$ such that $\{u, v\} \in W_x$ and as such, $V(H^{(v)}) \cap V(H^{(u)}) \neq \emptyset$. Hence $H[\{t : v \in B_t\}]$ is a connected subgraph of H for all $v \in V(G^2)$. We conclude that $(H, \mathcal{B})$ is indeed an H -decomposition of G^2 . Note that the width of $(H, \mathcal{B})$ is at most $(k + 1)(\Delta + 1) - 1$ since for every vertex in a bag of $(H, \mathcal{W})$, we add at most Δ other vertices to that bag to obtain $(H, \mathcal{B})$. $\square$

The next lemma is the heart of our characterisation of when a direct product has bounded treewidth or when it has bounded pathwidth. For the sake of simplicity, we shall consider a path to be a subdivision of K_1 .

Lemma 46. *For all connected graphs G_1, G_2 and H , where $\tau(G_1) = t$, and G_2 has maximum degree Δ and has an H -decomposition with width at most k , there exists an H' -decomposition of $G_1 \times G_2$ with width at most $t(k + 1)(\Delta + 1)$ where H' is a subdivision of H .*

Proof. Let $A \subseteq V(G_1)$ be a vertex cover of G_1 where $|A| = t$. We may assume that A is a clique in G_1 . Let $\tilde{G}_1 = G[A]$ and let $L := V(G_1) - A$. By Lemma 45, there exists an H -decomposition of G_2^2 with width at most $(k + 1)(\Delta + 1) - 1$. By Lemma 20, there exists an H -decomposition $(H, \mathcal{W})$ of $\tilde{G}_1 \boxtimes G_2^2$ that has width at most $t(k + 1)(\Delta + 1) - 1$.

If $L = \emptyset$, then $G_1 \boxtimes G_2 \subseteq \tilde{G}_1 \times G_2^2$ and we are done. Otherwise, let $(\ell, v) \in L \times V(G_2)$. Now $N_{G_1}(\ell)$ is a clique in K_t and $N_{G_2}(v)$ is a clique in G_2^2 . Thus,

$N_{G_1 \times G_2}((\ell, v))$ is a clique in $\tilde{G}_1 \boxtimes G_2^2$. As such, there exists a node $x \in V(H)$ such that $N_{G_1 \times G_2}((\ell, v)) \subseteq W_x$. We now explain how to modify $(H, \mathcal{W})$ to obtain an H' -decomposition $(H', \mathcal{B})$ of $G_1 \times G_2$ where H' is a subdivision of H .

Let $(\ell_1, v_1), (\ell_2, v_2), \dots, (\ell_j, v_j)$ be an arbitrary ordering of the vertices in $L \times V(G_2)$ where $j = |L \times V(G_2)|$. For each $i \in [j]$, let $x_i \in V(H)$ be a vertex where $N_{G_1 \times G_2}((\ell, v)) \subseteq W_{x_i}$. Initialise $i := 1, H_i := H, V_i := V(\tilde{G}_1 \boxtimes G_2^2)$ and $\mathcal{W}_i := \mathcal{W}$. For $i = 1, 2, \dots, j+1$, apply the following procedure: let y_i be a neighbour of x_i in H_i . Subdivide the edge $x_i y_i \in E(H_i)$ and let H_{i+1} be the graph obtained and let z_i be the new vertex from the subdivided edge (see Figure 3). Note that if no such $x_i y_i$ edge exists then $H_i = K_1$. In which case, let H_{i+1} be obtained from H_i by adding the vertex z_i and edge $x_i z_i$. Let $W_{z_i} = W_{x_i} \cup \{(\ell_i, v_i)\}$ and $\mathcal{W}_{i+1} = \mathcal{W}_i \cup \{W_{z_i}\}$ and $V_{i+1} = V_i \cup \{(\ell_i, v_i)\}$.

Once the above procedure has completed, let $H' := H_{j+1}$ and for all $x \in V(H')$, let $B_x := W_x$ and $\mathcal{B} := \{B_x : x \in V(H')\}$. Note that H' is a subdivision of H since H_{i+1} is obtained by subdividing an edge of H_i for all $i \in [j]$.

We now demonstrate that $(H', \mathcal{B})$ is an H' -decomposition of $G_1 \times G_2$. Let $(u_1, u_2) \in V(G_1 \times G_2)$. If $u_1 \in V(G_1) - L$, then $(u_1, u_2) \in V(\tilde{G}_1 \boxtimes G_2^2)$ and hence there is a $x \in V(H)$ such that $(u_1, u_2) \in W_x$. By the construction of $(H', \mathcal{B})$, it follows that $(u_1, u_2) \in B_x$. Otherwise, $u_1 \in L$ in which case $(u_1, u_2) = (\ell_i, v_i)$ for some $i \in [j]$ and $(u_1, u_2) \in W_{z_i}$. Hence every vertex is in a bag.

Let $(u_1, u_2)(u_3, v_4) \in E(G_1 \times G_2)$. Since L is an independent set, at most one of u_1 and u_3 is in L . Suppose that $u_1 \in L$. Then there exists $i \in [j]$ such that $(u_1, u_2) = (\ell_i, v_i)$. Since $(u_3, u_4) \in N_{G_1 \times G_2}[(\ell_i, v_i)]$, by the construction of B_{z_i} it follows that $(u_1, u_2), (u_3, v_4) \in B_{z_i}$. Now if neither u_1 nor u_3 is in L , then there is a node $x \in V(H)$ such that $(u_1, u_2), (u_3, u_4) \in W_x$ and hence $(u_1, u_2), (u_3, v_4) \in B_x$. Hence, the endpoints of each edge is in a bag.

It remains to show that for all $(u_1, u_2) \in V(G_1 \times G_2)$, the subgraph $H'[\{x \in V(H') : (u_1, u_2) \in B_x\}]$ is connected. We prove the following claim by induction on i :

Claim: For every vertex $(u_1, u_2) \in V_i$, the induced subgraph $H_i[\{x \in V(H') : (u_1, u_2) \in W_x\}]$ is connected.

For $i = 1$, this claim holds since $(H_1, \mathcal{W}_1)$ is an H -decomposition of $\tilde{G}_1 \boxtimes G_2^2$. Now assume that it holds up to $i - 1$ where $i \geq 2$. The graph H_i is obtained from H_{i-1} by replacing the edge $x_i y_i$ by the path x_i, z_i, y_i . Furthermore, $V_i \setminus V_{i-1} = \{(\ell_i, v_i)\}$. Now the

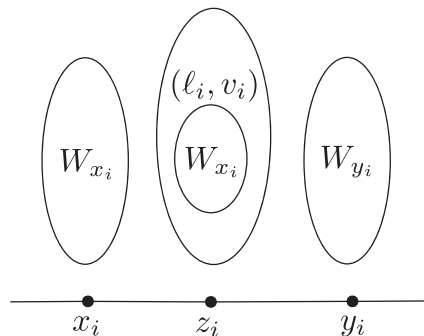


FIGURE 3 Subdividing the $x_i y_i$ edge in H_i to obtain H_{i+1} .

vertex (ℓ_i, v_i) is only in the bag W_{z_i} and hence $H_i[x : (\ell_i, v_i) \in W_x]$ is connected. Now if $(u_1, u_2) \in V_{i-1}$ then by induction $H_{i-1}[x : (u_1, u_2) \in W_x]$ is connected. For the sake of contradiction, suppose that $H_i[x : (u_1, u_2) \in W_x]$ is disconnected. The only way for this to occur is if $(u_1, u_2) \in W_{x_i}$ and $(u_1, u_2) \notin W_{z_i}$. However, by the construction of W_{z_i} , we have $W_{x_i} \subseteq W_{z_i}$, a contradiction. Hence $H_i[\{t : (u_1, u_2) \in W_t\}]$ is connected for all $(u_1, u_2) \in V_i$.

This completes our proof that $(H', \mathcal{B})$ is an H' -decomposition of $G_1 \times G_2$, where H' is a subdivision of H . We conclude by noting that the width of this H' -decomposition is at most 1 more than the width of $(H, \mathcal{W})$ as required. $\square$

When the graph H in Lemma 46 realises the treewidth or pathwidth of G_2 , we have the following.

Corollary 47. *For all connected graphs G_1 and G_2 ,*

$$\begin{aligned} \text{tw}(G_1 \times G_2) &\leq \tau(G_1)(\text{tw}(G_2) + 1)(\Delta(G_2) + 1), \text{ and} \\ \text{pw}(G_1 \times G_2) &\leq \tau(G_1)(\text{pw}(G_2) + 1)(\Delta(G_2) + 1). \end{aligned}$$

The next two lemmas are key cases for when a direct product has unbounded treewidth.

Lemma 48. *For every integer $k \geq 1$ and graph G with $\text{dll}(G) \geq k$, we have $K_{k,k}$ is a minor of $G \times P_{2k}$ and thus $\text{tw}(G \times P_{2k}) \geq k$.*

Proof. Since $\text{dll}(G) \geq k$, it follows that G contains a tree T such the subtree T_L obtained by deleting the leaves set of T has at least k leaves. We claim that $T \times P_{2k}$ contains $K_{k,k}$ as a minor. Since $T \times P_{2k}$ is a subgraph of $G \times P_{2k}$, this implies that $K_{k,k}$ is also a minor of $G \times P_{2k}$.

Let $P_{2k} = (p_1, \dots, p_{2k})$. Let $M = \{u_1 v_1, u_2 v_2, \dots, u_k v_k\}$ be a matching in T such that for each $i \in [k]$, u_i is a leaf of T_L and v_i is a leaf of T . Let T' be the subtree of T_L obtained by deleting $u_1, \dots, u_k$. Take a proper black-white colouring of $V(T)$. Let B be the set of black vertices and let W be the set of white vertices. For each $j \in [k]$, let $x_j \in \{u_j, v_j\} \cap B$ and $y_j \in \{u_j, v_j\} \cap W$. Let X_j be the subgraph of $T \times P_{2k}$ induced by

$$\{(x_j, p_i), (y_j, p_{i-1}) : i \in \{2, 4, \dots, 2k\}\}.$$

Since $u_j v_j \in E(T)$ and P_{2j} is bipartite, it follows that X_j is a copy of P_{2j} . Moreover, X_i and X_j are vertex-disjoint whenever $i \neq j$ since M is a matching.

For each $i \in [k]$, let Y_i be the subgraph of $T \times P_{2k}$ induced by

$$\{(x, p_{2i}) : x \in V(T') \cap B\} \cup \{(y, p_{2i-1}) : y \in V(T') \cap W\}.$$

Since $p_{2i-1} p_{2i} \in E(P_{2k})$ and T' is bipartite, it follows that Y_i is a copy of T' . Moreover, since $\{p_{2i}, p_{2i-1}\} \cap \{p_{2j}, p_{2j-1}\} = \emptyset$ whenever $i \neq j$, it follows that Y_i and Y_j are vertex-disjoint.

We claim that $(\{X_j : j \in [k]\}, \{Y_i : i \in [k]\})$ is a $K_{k,k}$ model in $T \times P_{2k}$. Since $V(T') \cap \{u_1, v_1, \dots, u_k, v_k\} = \emptyset$, it follows the X_j and Y_i 's are pairwise vertex-disjoint. It

therefore remains to show that X_j and Y_i are adjacent. For each $j \in [k]$, let $z_j \in V(T')$ be the neighbour of u_j . Then for all $i, j \in [k]$, either the edge $(u_j, p_{2i-1})(z_j, p_{2i})$ or the edge $(u_j, p_{2i})(z_j, p_{2i-1})$ is between X_i and Y_j (depending on whether u_i is white or black). In either case, X_i and Y_j are adjacent which completes the proof for the claim that $K_{k,k}$ is a minor of $T \times P_{2k}$. $\square$

The next lemma is folklore and uses the well-known fact that $\text{tw}(P_n \square P_n) = n$ [5].

Lemma 49. *For integers $b, n \geq 1$, let S_b be the star with b leaves and P_n be the path on n vertices. Then $\text{tw}(S_b \times S_b) \geq b$ and $\text{tw}(P_{2n-1} \times P_{2n-1}) \geq n$.*

Proof. For the direct product of stars, $K_{b,b} \subseteq S_b \times S_b$ by Theorem 6 and hence $\text{tw}(S_b \times S_b) \geq b$.

Now consider the direct product of two paths. Since $\text{tw}(P_n \square P_n) \geq n$, it suffice to show that $P_n \square P_n \subseteq P_{2n-1} \times P_{2n-1}$. Let $V(P_{2n-1}) = [2n-1]$. For each $i \in [n]$, let $H^{(i)}$ be the subgraph of $P_{2n-1} \times P_{2n-1}$ induced by $\{(i-k, n+i-k) : k \in [n]\}$. For each $j \in [n]$, let $P^{(j)}$ be the subgraph of $P_{2n-1} \times P_{2n-1}$ induced by $\{(j+k, n-j+k) : k \in [n]\}$. Then $(\bigcup_{i \in [n]} H^{(i)}) \cup (\bigcup_{j \in [n]} P^{(j)})$ defines a $P_n \square P_n$ subgraph in $P_{2n-1} \times P_{2n-1}$ where for all $i, j \in [n]$, $H^{(i)}$ is the i^{th} -horizontal path in $P_n \square P_n$ and $P^{(j)}$ is the j^{th} -vertical path in $P_n \square P_n$, as required (see Figure 4). $\square$

The next lemma is the Moore bound; see [53] for a survey.

Lemma 50. *For every connected graph G with maximum degree $\Delta > 1$ and diameter d ,*

$$v(G) \leq \begin{cases} 1 + \Delta \frac{(\Delta-1)^d - 1}{\Delta-2}, & \text{if } \Delta > 2, \\ 2d+1, & \text{if } \Delta = 2. \end{cases}$$

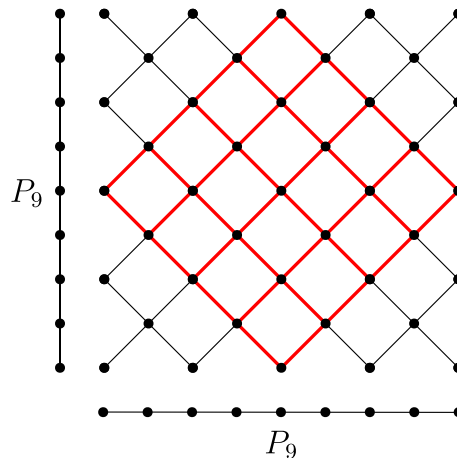


FIGURE 4 The (5×5) -grid in $P_9 \times P_9$.

For a graph G , let $\tilde{\tau}(G)$ be the maximum vertex cover number of a component of G . We now prove our characterisation for when a direct product has bounded treewidth.

Theorem 51. *Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be monotone graph classes that contains K_2 . Then $\text{tw}(\mathcal{G}_1 \times \mathcal{G}_2)$ is bounded if and only if $\text{tw}(\mathcal{G}_1)$ and $\text{tw}(\mathcal{G}_2)$ are bounded and at least one of the following conditions hold:*

- $\tilde{v}(\mathcal{G}_1)$ or $\tilde{v}(\mathcal{G}_2)$ is bounded;
- $\tilde{\tau}(\mathcal{G}_1)$ and $\Delta(\mathcal{G}_2)$ are bounded; or
- $\tilde{\tau}(\mathcal{G}_2)$ and $\Delta(\mathcal{G}_1)$ are bounded.

Proof. Assume that $\text{tw}(\mathcal{G}_1 \times \mathcal{G}_2)$ is bounded. By Corollary 31, if $\mathcal{G}_1$ or $\mathcal{G}_2$ has unbounded treewidth then $\mathcal{G}_1 \times \mathcal{G}_2$ has unbounded treewidth. Hence, we may assume that there exists an integer $k \geq 1$ such that $\text{tw}(\mathcal{G}_1) \leq k$ and $\text{tw}(\mathcal{G}_2) \leq k$. Now suppose there exists an integer $c_1 \geq 1$ such that $\tilde{v}(\mathcal{G}_1) \leq c_1$ or $\tilde{v}(\mathcal{G}_2) \leq c_1$. By Lemma 19, $\text{tw}(\mathcal{G}_1 \times \mathcal{G}_2) \leq (k+1)c_1 - 1$ and we are done.

It remains to consider the case when neither $\tilde{v}(\mathcal{G}_1)$ nor $\tilde{v}(\mathcal{G}_2)$ is bounded. Suppose that $\mathcal{G}_1$ and $\mathcal{G}_2$ both have unbounded maximum degree. Then for every integer $b \geq 1$, the star S_b is a member of both $\mathcal{G}_1$ and $\mathcal{G}_2$ and $(S_b \times S_b)_{b \in \mathbb{N}} \subseteq \mathcal{G}_1 \times \mathcal{G}_2$. However, by Lemma 49, this is a contradiction since $(S_b \times S_b)_{b \in \mathbb{N}}$ has unbounded treewidth. Therefore either $\mathcal{G}_1$ or $\mathcal{G}_2$ has bounded maximum degree.

Without loss of generality, there exists an integer $c_2 \geq 1$ such that $\Delta(\mathcal{G}_2) \leq c_2$. Since $\tilde{v}(\mathcal{G}_2)$ is unbounded, by Lemma 50 it follows that $\text{path}(\mathcal{G}_2)$ is unbounded. Now if $\text{path}(\mathcal{G}_1)$ is also unbounded, then $(P_n \times P_n)_{n \in \mathbb{N}}$ is a family of graphs in $\mathcal{G}_1 \times \mathcal{G}_2$. This contradicts $\text{tw}(\mathcal{G}_1 \times \mathcal{G}_2)$ being bounded since by Lemma 49, $(P_n \times P_n)_{n \in \mathbb{N}}$ has unbounded treewidth. Similarly, if $\text{dll}(\mathcal{G}_1)$ is unbounded, then by Lemma 48, $\text{tw}(\mathcal{G}_1 \times \mathcal{G}_2)$ is unbounded. So assume there exists integers $j, n \geq 1$ such that for every connected graph $G \in \mathcal{G}_1$, we have $\text{dll}(G) \leq j$ and $\text{path}(G) \leq n$. By Lemma 44, $\hat{\tau}(\mathcal{G}_1) \leq \lceil (j+1)n/2 \rceil$. By Lemma 47, $\text{tw}(\mathcal{G}_1 \times \mathcal{G}_2) \leq \lceil (j+1)n/2 \rceil (k+1)(\Delta+1)$ and thus, is bounded. As we have considered all possibilities, this completes our proof. $\square$

The next theorem characterises when a direct product has bounded pathwidth. We omit the proof as it is identical to Theorem 51 except we use Corollary 33 instead of Corollary 31.

Theorem 52. *Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be monotone graph classes that contains K_2 . Then $\mathcal{G}_1 \times \mathcal{G}_2$ has bounded pathwidth if and only if $\mathcal{G}_1$ and $\mathcal{G}_2$ both have bounded pathwidth and at least one of the following holds:*

- $v(\hat{\mathcal{G}}_1)$ or $v(\hat{\mathcal{G}}_2)$ is bounded;
- $\tau(\hat{\mathcal{G}}_1)$ and $\Delta(\mathcal{G}_2)$ are bounded; or
- $\tau(\hat{\mathcal{G}}_2)$ and $\Delta(\mathcal{G}_1)$ are bounded.

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Data sharing not applicable to this article as no data sets were generated or analysed during the current study.

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